

B.V.Sc & A.H — 1st PROFESSIONAL

OnlyVet Anatomy Notes

Volume 2 — Myology

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for OnlyVet — in the language of love, care and compassion

Points + Tables + Bold Keywords + PYQs — Exam-Focused Format

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Myology — Overview & Classification

Definition

- **Myology** = study of muscles (musculi) + their accessory structures (fasciae, synovial membranes)
- Muscles = highly specialized organs; property of **contractibility** (contract under stimulus) → moves body parts
- Muscles = the **active part** of locomotion (bones = passive part)
- **Sylvius** = anthropotomist who first named the muscles

Classification — By Structure (3 Types)

Muscle Type	Also Called
Skeletal muscle	Striated / Voluntary
Smooth muscle	Non-striated / Involuntary / Visceral
Cardiac muscle	Involuntary + Striated (unique combination)

Classification — By Function (2 Types)

- **Voluntary muscle** (controlled by will — skeletal)
- **Involuntary muscle** (not controlled by will — smooth + cardiac)

IMPORTANT

Cardiac muscle is the **ONLY** type that is **BOTH** striated **AND** involuntary — a frequently tested exception to the usual "striated=voluntary" rule.

Accessory Structures of Muscle

Structure	Subtypes
Fascia	Superficial fascia, Deep fascia
Synovial membranes	Synovial bursa, Tendon sheath

PYQ

Which muscle type is both striated and involuntary?

A) Skeletal B) Smooth **C) Cardiac ✓** D) None

Exp: Cardiac muscle uniquely combines striated appearance with involuntary control.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Classify muscles based on structure and function, giving examples.

■ Quick Recap

- Myology = muscles + fascia + synovial membranes
- 3 structural types: Skeletal (striated/voluntary), Smooth (non-striated/involuntary), Cardiac (striated/involuntary)
- Accessory structures: Fascia (superficial/deep) + Synovial membranes (bursa/tendon sheath)

Skeletal/striated Muscle — Structure

General Features

- Directly/indirectly attached to skeleton → "skeletal muscles"
- Long, unbranched fibres with cross striations (microscopic) → "striated muscle"
- Contraction = **per the animal's will** → "voluntary muscles"
- Each muscle = an individual organ, made of several fibres
- **~200-250 paired + few unpaired muscles** in domestic mammals

Connective Tissue Sheaths (Fibre → Whole Muscle)

Level	Sheath Name
Single muscle fibre	Endomysium
Group of fibres (fasciculus)	Perimysium
Whole muscle (many fasciculi)	Epimysium (associated with/sometimes fused to fascia)

MNEMONIC

"Endo-Peri-Epi" small to large — Endomysium (1 fibre) → Perimysium (fasciculus) → Epimysium (whole muscle).

Muscle Parts

- Central portion = **belly**; 2 ends attach to bone/cartilage/skin via **tendon** or **ligament**
- On contraction: one end stays **FIXED (origin)**, other end **MOVES (insertion)**
- In limbs: distal part is usually the mobile one → distal attachment = insertion (general rule)

Tendon vs Aponeurosis

Structure	Description	Example
Tendon	Narrow band of white fibrous tissue, muscle→bone/cartilage/tissue	—
Aponeurosis	Broad SHEET of fibrous connective tissue, often glistening	Linea alba

Blood & Nerve Supply

- Muscle tissue = richly supplied with blood; large arteries accompanied by veins + lymphatics
- Lymph vessels = **FEW**
- Nerve fibre → muscle fibres → ramified expansions = **end plates**
- Muscle nerves = **sensory + vasomotor**

Naming Conventions for Voluntary Muscles (6 Criteria)

Naming Criterion	Explanation	Example
Colour	Red (deep, sustained/continuous work, high myoglobin/cytochrome) vs White (superficial, quick/intermittent work)	—
Action/Force	Spurt muscles (acceleration motion at joint) vs Shunt muscles (stabilizing centripetal force); also named by action: flexor/extensor	Brachialis (spurt); Flexor carpi radialis (shunt + flexor); Extensor carpi radialis (extensor)

Naming Criterion	Explanation	Example
Shape	Named for distinctive shape	Deltoides ("deltoid" = triangular)
Location	Named for body region	Biceps brachii (brachial/arm region)
Direction of fibres	See table below	—
Number of heads/divisions	Number of origin attachment sites	Biceps brachii (2 heads)
Attachment sites	Origin + insertion combined in name	Sterno-cephalicus (origin=sternum, insertion=head/cephalic)

Direction of Fibres — 5 Sub-Types

Fibre Arrangement	Description	Example
Parallel	Long fibres, parallel to line of pull	Biceps brachii
Pennate	Short fibres, attached to tendon at acute angle (uni/bi/multipennate by fibre direction relative to tendon)	Deltoid
Spiral	Twisted fibre arrangement — helps rotational movement	Brachialis
Cruciate	Fibres cross like "X" in superficial + deep planes	Masseter
Rectus / Oblique	Rectus = straight lengthwise fibres; Oblique = fibres run at an angle	Rectus abdominis / Obliquus abdominis

EXAM TIP

"Pennate" comes from Latin for "feather" — fibres attach to the tendon like feather barbs to a shaft, at an acute angle, unlike parallel muscles which run straight along the pull line.

PYQ

Muscle fibres arranged in a twisted manner, helping rotational movement, are called?

A) Parallel B) Pennate **C) Spiral ✓** D) Cruciate

Exp: Spiral muscle fibres (e.g. brachialis) are twisted, aiding rotational movements.

PYQ

The endomysium surrounds which structure?

A) A single muscle fibre ✓ B) A fasciculus C) The whole muscle D) A tendon

Exp: Endomysium surrounds each individual muscle fibre; perimysium covers the fasciculus; epimysium covers the whole muscle.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the six criteria used for naming voluntary muscles, with one example each.

Quick Recap

- Endomysium (fibre) → Perimysium (fasciculus) → Epimysium (whole muscle)
- Muscle = Belly + 2 ends (Origin=fixed, Insertion=movable, usually distal in limbs)
- Tendon = narrow band | Aponeurosis = broad sheet (e.g. linea alba)
- Naming: Colour, Action, Shape, Location, Fibre direction, Number of heads, Attachment sites
- Fibre types: Parallel, Pennate (uni/bi/multi), Spiral, Cruciate, Rectus/Oblique

MYOLOGY — INTRODUCTION

Smooth Muscle

- Also called: **Non-striated** or **Involuntary** muscle (contraction NOT under animal's will)
- No cross striations under microscope → "smooth muscle"
- Forms the bulk of visceral organ walls → also called **visceral muscle**
- Composed of **fusiform/spindle-shaped cells**, single nucleus at center
- Fibres arranged generally **parallel** to each other

PYQ

Smooth muscle is composed of which cell shape?

A) Branched cylindrical **B) Fusiform/spindle-shaped ✓** C) Cuboidal D) Flat squamous

Exp: Smooth muscle cells are fusiform (spindle-shaped) with a single central nucleus.

■ Quick Recap

- Smooth muscle = non-striated, involuntary, visceral muscle; fusiform cells, single central nucleus

MYOLOGY — INTRODUCTION

Cardiac Muscle

- Found **ONLY** in: the heart + immediate proximal ends of aorta, pulmonary artery, pulmonary veins
- = **Involuntary AND Striated** (unique combination — not under animal's control, yet shows cross striations like skeletal muscle)
- Unlike skeletal fibres: **single nucleated**, smaller, often **multiple branches**
- Cells attach to adjacent cells forming a **branching network**
- Firm end-to-end attachments between cells = visible dark transverse lines = **intercalated discs**

IMPORTANT

Intercalated discs are the defining microscopic feature distinguishing cardiac muscle from skeletal muscle under the microscope — both are striated, but only cardiac has these discs.

PYQ

The dark transverse lines marking firm end-to-end attachments between cardiac muscle cells are called?

A) Endomysium **B) Intercalated discs ✓** C) Sarcomeres D) Z-lines

Exp: Intercalated discs are the visible junctions between adjacent cardiac muscle cells.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Compare skeletal, smooth, and cardiac muscle in terms of structure and control.

■ Quick Recap

- Cardiac muscle = involuntary + striated (unique); found only in heart + great vessel roots
- Single nucleus, branched, joined by intercalated discs

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Muscles of the Pectoral Girdle — Pectoral Muscles

Overview

- Forelimb attaches to body via **muscles of the shoulder girdle** — connect forelimb to head, neck, trunk
- List: Pectoral muscles (Superficial: Cranial + Caudal; Deep) | Trapezius (cervicis + thoracis) | Brachiocephalicus | Omotransversarius | Latissimus dorsi | Serratus ventralis (cervicis + thoracis) | Rhomboideus (cervicis + thoracis)

Axilla

- Narrow space between chest wall and pectoral limb, filled with loose connective tissue
- Crossed by large vessels + nerves supplying the pectoral limb muscles
- Corresponds to the human **armpit**
- Barely exists at rest (small extent, lower part); clearly seen when limb is **abducted**

Pectoral Muscles — 2 Layers (Superficial + Deep)

Muscle	Origin	Insertion	Action
Anterior superficial pectoral	Ventro-lateral aspect, anterior sternum	Common with brachiocephalicus, anterior edge of humerus	Adduct limb, draw it forward
Posterior superficial pectoral	Ventral/lateral anterior sternum, behind the preceding	Crest of humerus + forearm fascia	Adduct limb, tense forearm fascia
Deep pectoral (Pectoralis profundus)	Ventral sternum surface + tunica abdominalis	Medial tuberosity of humerus	Adduct AND retract the limb

- Deep pectoral: large, fleshy; extends xiphoid cartilage level → shoulder joint level
- Deep pectoral — Blood: External thoracic artery | Nerve: Pectoral nerves

EXAM TIP

Superficial pectorals = adduct only; Deep pectoral = adduct AND retract — the extra retraction action is what distinguishes the deep layer.

PYQ

Which pectoral muscle layer is responsible for both adduction and retraction of the limb?

A) Anterior superficial pectoral B) Posterior superficial pectoral **C) Deep pectoral ✓** D) None

Exp: The deep pectoral (pectoralis profundus) both adducts and retracts the limb, unlike the superficial layers which only adduct.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the pectoral muscles under superficial and deep layers with their actions.

■ Quick Recap

- Axilla = armpit equivalent, seen clearly on limb abduction
- Superficial pectorals (ant+post) = adduct only; Deep pectoral = adduct + retract

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Trapezius, Brachiocephalicus & Omotransversarius

Trapezius

- Broad triangular muscle; dorsal midline, atlas level → end of thoracic region; covers part of shoulder
- 2 parts: **Cervical (trapezius cervicis)** + **Dorsal (trapezius thoracis)**

Feature	Detail
Origin	Ligamentum nuchae + supraspinous ligament, atlas level to T12
Insertion	Tuberos part of scapular spine + scapular fascia
Action	Elevate shoulder as a whole; cervical part draws forward, dorsal part draws backward
Blood supply	Deep cervical, dorsal, intercostal arteries
Nerve supply	Spinal accessory nerve

Brachiocephalicus

- Extends head → along neck → arm; upper border apposed to trapezius; lower border = upper boundary of jugular furrow
- 2 parts: **Dorsal (cleido-occipitalis)** + **Ventral (cleido-mastoideus)**

Feature	Detail
Origin	Dorsal part: occipital bone, ligamentum nuchae. Ventral part: mastoid process, atlas wing, mandible
Insertion	United portion + anterior superficial pectoral → anterior edge of humerus
Action	Head/neck fixed: draws limb forward, extends shoulder. Limb fixed: both sides extend head/neck; single side inclines head/neck to its own side
Note	Deep face receives a small red fasciculus from 1st rib cartilage — vestige of man's subclavius
Blood supply	Inferior cervical, carotid, vertebral arteries
Nerve supply	Cervical spinal + axillary nerves

Omotransversarius

- Atlas level → shoulder; mostly covered by brachiocephalicus except at scapular portion (broad flat band there)

Feature	Detail
Origin	Wing of atlas + transverse process of axis
Insertion	Scapular spine + scapular fascia
Action	Pull lower scapular angle forward-upward; tense scapular fascia
Blood supply	Superior + inferior cervical arteries
Nerve supply	Cervical spinal nerves

IMPORTANT

Brachiocephalicus's "vestige of subclavius" fasciculus is a classic comparative-anatomy fact linking veterinary anatomy to human anatomy.

PYQ

The brachiocephalicus muscle consists of which two parts?

- A) Cervical and thoracic **B) Dorsal (cleido-occipitalis) and ventral (cleido-mastoideus) ✓** C) Superficial and deep D) Medial and lateral

Exp: The brachiocephalicus has a dorsal part (cleido-occipitalis) and ventral part (cleido-mastoideus).

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the origin, insertion, and action of the trapezius muscle.

■ Quick Recap

- Trapezius: elevates shoulder (cervical=forward, dorsal=backward); nerve=spinal accessory
- Brachiocephalicus: head/neck-limb interaction muscle; has "vestige of subclavius"
- Omotransversarius: pulls scapula's lower angle forward-upward

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Latissimus Dorsi & Serratus Ventralis

Latissimus Dorsi

- Very wide muscle; lateral thorax wall, dorso-lumbar spine → arm; aponeurotic origin → wide muscular portion

Feature	Detail
Origin	Lumbodorsal fascia, 11th & 12th ribs, fascia over intercostal + external oblique muscles
Insertion	Anterior part (with teres major) → medial/teres tubercle of humerus; Middle part → aponeurosis on triceps long head; Posterior part (with deep pectoral) → medial tuberosity of humerus. Tensor fasciae antibrachii originates from its lateral surface
Action	Flex the shoulder joint
Blood supply	Thoraco-dorsal artery
Nerve supply	Thoraco-dorsal nerve

Serratus Ventralis — 2 Parts

- Large fleshy muscle, lateral neck + thorax; named **serratus cervicis** or **serratus thoracis** by region covered

Part	Detail
Serratus cervicis	Large, thick; C2 level → 5th rib. Origin: C2/C3 transverse process → C7, + ribs 1-5 lateral face. Insertion: triangular rough area, dorso-anterior ventral scapula surface. Action: pulls scapula toward neck (together=extend neck; singly=incline neck to own side). Blood: superior cervical, dorsal, vertebral arteries. Nerve: C4,5,6 spinal + long thoracic nerve
Serratus thoracis	Thin, flat, strong glistening aponeurosis; overlaps serratus cervicis; ventral border serrated, 6 digitations. Origin: ribs 4-9 external surfaces. Insertion: rough line, dorso-posterior ventral scapula surface. Action: BOTH sides = sling suspending trunk between forelimbs; SINGLY = pulls scapula up-back; limb fixed = extra-ordinary inspiratory muscle. Blood: intercostal arteries. Nerve: long thoracic nerve

IMPORTANT

Serratus thoracis + cervicis together form the "sling" that suspends the entire trunk between the two forelimbs — a key functional/biomechanical fact (this is essentially the synsarcosis concept in action).

PYQ

The muscle that acts as a "sling" suspending the trunk between the two forelimbs is the?

A) Trapezius B) **Serratus ventralis ✓** C) Rhomboideus D) Latissimus dorsi

Exp: Serratus ventralis (cervicis + thoracis parts together) forms the muscular sling suspending the trunk between the forelimbs.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the serratus ventralis muscle and its two parts with their actions.

■ Quick Recap

- Latissimus dorsi: flexes shoulder joint; shares insertion with teres major + deep pectoral
- Serratus ventralis = cervicis + thoracis; together = trunk-suspending sling; singly = scapula mover; also aids forced inspiration

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Rhomboideus

- Extends T9-T10 → mid funicular part of ligamentum nuchae; under cover of trapezius
- 2 parts: **Rhomboideus cervicis** + **Rhomboideus thoracis**

Feature	Detail
Origin	Ligamentum nuchae + spines of dorsal vertebrae (T1 to about T7/T8)
Insertion	Medial surface of the scapular cartilage
Action	Draw the shoulder upward and forward
Blood supply	Superior cervical + dorsal arteries
Nerve supply	C5, C6, C7, C8 spinal nerves

PYQ

The rhomboideus muscle inserts onto which structure?

A) Deltoid tuberosity B) **Medial surface of scapular cartilage ✓** C) Olecranon D) Humeral crest

Exp: The rhomboideus inserts onto the medial surface of the cartilage of the scapula.

■ Quick Recap

- Rhomboideus (cervicis + thoracis): draws shoulder up-forward; under trapezius

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Muscles of Shoulder — Medial Group

Overview

- Shoulder muscles = **Medial group** (Teres major, Subscapularis, Coracobrachialis) + **Lateral group** (Deltoides, Supraspinatus, Infraspinatus, Teres minor)

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Teres major	Dorsal angle of scapula + adjacent posterior border	Short flat tendon (with latissimus dorsi) → medial tubercle of humerus	Flex shoulder, adduct arm	Subscapular artery	Nerve to teres major
Subscapularis	Subscapular fossa (3 fleshy parts)	Flat tendon → posterior division, medial tuberosity of humerus (covered by coracobrachialis tendon)	Adduct the arm	Subscapular artery	Nerves to subscapularis
Coracobrachialis	Coracoid process of scapula	Small area, upper medial humerus (above teres tubercle) + middle 1/3 anterior humeral surface	Adduct arm, flex shoulder joint	Anterior circumflex artery	Musculo-cutaneous nerve

EXAM TIP

All 3 medial group muscles ADDUCT the arm — this is their shared functional theme, contrasting with the lateral group's abduction role.

PYQ

Which medial group shoulder muscle originates from the coracoid process of the scapula?

A) Teres major B) Subscapularis **C) Coracobrachialis ✓** D) Deltoideus

Exp: Coracobrachialis originates from the coracoid process, as its name suggests (coraco- = coracoid, brachialis = arm).

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the medial group muscles of the shoulder with their actions.

■ Quick Recap

- Medial group: Teres major, Subscapularis, Coracobrachialis — ALL adduct the arm

MYOLOGY — PECTORAL GIRDLE & SHOULDER

Muscles of Shoulder — Lateral Group

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Deltoideus	Acromial part: acromion process + scapular fascia. Scapular part: posterior scapular border	Deltoid tuberosity of humerus (scapular part mostly → triceps fascia)	Flex shoulder, abduct arm	Posterior circumflex artery	Axillary nerve
Supraspinatus	Supraspinous fossa (extends beyond it, contacts subscapularis)	Divides below acromion into medial + lateral divisions on humeral tuberosities	Extend the shoulder	Suprascapular artery	Suprascapular nerve

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Infraspinatus	Infraspinous fossa (extends beyond posteriorly)	2 insertions: circular area below lateral tuberosity summit + medial convexity surface	Abduct scapula, rotate it outward	Subscapular + suprascapular arteries	Suprascapular nerve
Teres minor	Rough lines, distal infraspinous fossa + posterior scapular border	Deltoid tuberosity + small rough area above it	Flex shoulder, abduct arm, rotate it outward	Subscapular + suprascapular arteries	Axillary nerve

EXAM TIP

All 4 lateral group muscles either ABDUCT or help with shoulder flexion/extension — the opposite functional theme to the medial group's pure adduction.

PYQ

Which lateral group shoulder muscle's action is specifically to extend the shoulder?

A) Deltoideus B) **Supraspinatus** ✓ C) Infraspinatus D) Teres minor

Exp: Supraspinatus's primary action is to extend the shoulder joint.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the deltoideus muscle with its two parts, origin, insertion, and action.

Quick Recap

- Lateral group: Deltoideus (flex+abduct), Supraspinatus (extend), Infraspinatus (abduct scapula+rotate out), Teres minor (flex+abduct+rotate out)
- Medial group = adductors; Lateral group = abductors/extensors — opposite functional roles

MYOLOGY — ARM, FOREARM & MANUS

Muscles of the Arm

At a Glance

Muscle	Primary Action	Nerve
Biceps brachii	Flex elbow	Musculo-cutaneous
Triceps brachii (3 heads)	Extend elbow (long head also flexes shoulder)	Radial
Tensor fascia antibrachii	Extend elbow, tense forearm fascia	Radial
Brachialis	Flex elbow	Musculo-cutaneous (median branch)
Anconeus	Extend elbow	Radial

EXAM TIP

This group has a clean split — Biceps + Brachialis = FLEXORS; Triceps + Tensor fascia antibrachii + Anconeus = EXTENSORS of the elbow. Memorize the split, not just individual muscles.

Biceps Brachii

- Long, fusiform muscle; anterior humeral face, passes obliquely down-back-in; thick at middle, **bifid at insertion**

Feature	Detail
Origin	Tuber scapulae
Insertion	Radial tuberosity + Ulna (bifid)
Action	Flex the elbow joint
Blood supply	Anterior circumflex + direct branch from brachial artery
Nerve supply	Musculo-cutaneous nerve

Triceps Brachii — 3 Heads

- Large muscle, fills the space between scapula and ulna's olecranon process

Head	Size/Shape	Origin	Insertion	Action
Long head (Caput longum)	LARGEST; visible on lateral + medial faces	Posterior scapular border + humerus	Lateral/posterior/medial parts of olecranon summit	Flex SHOULDER + extend elbow (only head with dual action)
Lateral head (Caput parvum)	Quadrilateral, lateral surface below long head	Curved line, deltoid tuberosity to humeral neck	Long head's tendon + olecranon's lateral aspect	Extend elbow
Medial head (Caput medium)	SMALLEST, medial surface below coracobrachialis insertion	Upper 1/3 medial humeral shaft	Anterior/medial olecranon summit	Extend elbow

- Blood (medial head): Posterior circumflex + deep brachial arteries | Nerve: Radial nerve

IMPORTANT

Only the LONG head of triceps crosses the shoulder joint — that's why it uniquely flexes the shoulder in addition to extending the elbow. Lateral/medial heads act on the elbow only.

Tensor Fascia Antibrachii, Brachialis & Anconeus

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Tensor fascia antibrachii	Latissimus dorsi	Tendon of triceps long head	Extend elbow, tense forearm fascia	Posterior circumflex + deep brachial arteries	Radial nerve
Brachialis	Posterior humeral surface (lodged in musculo-spiral groove)	Upper medial radius (below bicipital tuberosity) + ulna	Flex elbow joint	Posterior circumflex + deep brachial arteries	Musculo-cutaneous branch of median nerve
Anconeus	Margins of olecranon fossa	Anterior + lateral olecranon	Extend elbow joint	Deep brachial + posterior circumflex arteries	Radial nerve

PYQ

Which head of the triceps brachii uniquely acts on both the shoulder and elbow joints?

A) Lateral head B) Medial head **C) Long head ✓** D) All three equally

Exp: Only the long head crosses the shoulder joint, so it flexes the shoulder in addition to extending the elbow.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the three heads of the triceps brachii with their origins and actions.

Quick Recap

- Flexors of elbow: Biceps brachii + Brachialis
- Extensors of elbow: Triceps (3 heads) + Tensor fascia antibrachii + Anconeus
- ONLY Triceps long head also flexes the shoulder (crosses 2 joints)

MYOLOGY — ARM, FOREARM & MANUS

Muscles of the Forearm — Extensors

At a Glance

- Location: dorsal + lateral aspects of forearm | Covered by superficial + deep fascia
- All 5 extensors share one blood supply (Collateral radial artery) and one nerve (Radial nerve)

Muscle	Primary Action
Extensor carpi radialis	Extend carpus, flex elbow (LARGEST extensor)
Medial digital extensor	Extend medial digit joints, abduct digit
Common digital extensor	Extend joints from carpus downward, approximate digits
Lateral digital extensor	Extend lateral digit joints, abduct them
Extensor carpi obliquus	Extend carpus, rotate it outward

Detail Table

Muscle	Origin	Insertion	Key Course Note
Extensor carpi radialis	Sharp ridge, lateral humeral epicondyle + rough pit outside coronoid fossa	Wide flat tendon to metacarpal tuberosity	LARGEST extensor; tendon crossed by extensor carpi obliquus tendon
Medial digital extensor	Lateral humeral epicondyle + rough pit outside coronoid fossa	2 branches to 2nd + 3rd phalanges of medial digit	Tendon receives 2 slips from suspensory ligament over 1st phalanx
Common digital extensor	Lateral humeral epicondyle + ulna (2 heads, fuse mid-forearm)	Extensor process of 3rd phalanx (both branches, at fetlock split)	Runs with medial digital extensor's tendon in common synovial sheath
Lateral digital extensor	Lateral elbow ligament, radius lateral tuberosity, radius/ulna shafts	2nd + 3rd phalanges of lateral digit	Receives 2 suspensory ligament slips at 1st phalanx
Extensor carpi obliquus	Lateral + dorsal radius/ulna faces	Tubercle, postero-medial upper large metacarpal	Crosses OVER extensor carpi radialis tendon obliquely inward

EXAM TIP

Every extensor tendon threading down to the digits picks up "slips from the suspensory ligament" near the fetlock — a repeated anatomical theme worth remembering as a pattern, not memorizing per-muscle.

PYQ

Which is the largest of the forearm extensor muscles?

A) Common digital extensor **B) Extensor carpi radialis ✓** C) Lateral digital extensor D) Extensor carpi obliquus
Exp: Extensor carpi radialis is the largest extensor, lying over the dorsal radius from humerus to the large metacarpal.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the extensor muscles of the forearm with their actions.

Quick Recap

- 5 extensors, all dorsal/lateral forearm, all supplied by Radial nerve + Collateral radial artery
- Extensor carpi radialis = largest; digital extensors (medial/common/lateral) act on the digits

MYOLOGY — ARM, FOREARM & MANUS

Muscles of the Forearm — Flexors

At a Glance — Two Layers

Layer	Muscles
Superficial	Pronator teres, Flexor carpi radialis (medial), Flexor carpi ulnaris (middle), Ulnaris lateralis
Deep	Superficial digital flexor, Deep digital flexor

- Location: posterior + medial forearm aspects

Superficial Layer

Muscle	Origin	Insertion	Action	Nerve
Pronator teres	Medial humeral epicondyle	Upper medial radial border	INAPPRECIABLE (vestigial!)	Median nerve
Flexor carpi radialis (Medial flexor of carpus)	Medial humeral epicondyle	Tubercle, postero-medial large metacarpal	Flex carpus, extend elbow	Median nerve
Flexor carpi ulnaris (Middle flexor of carpus)	2 heads: humeral (medial epicondyle) + ulnar (olecranon medial face)	Supero-medial half of accessory carpal	Flex carpus, extend elbow	Ulnar nerve
Ulnaris lateralis (Flexor metacarpi externus)	Lateral humeral epicondyle	Short tendon to supero-lateral carpal bone + long tendon to lateral large metacarpal	Flex carpus, extend elbow	Radial nerve

IMPORTANT

Pronator teres is VESTIGIAL in these species — its action is "inappreciable." A classic trick-question muscle: it exists anatomically but does essentially nothing functionally.

Deep Layer

Muscle	Heads of Origin	Insertion	Action	Nerve
Superficial digital flexor (Flexor pedis perforatus)	Medial humeral epicondyle (single origin, belly divides into 2 parts)	Volar face, upper 2nd phalanges	Flex digits + carpus, extend elbow	Ulnar + median nerves

Muscle	Heads of Origin	Insertion	Action	Nerve
Deep digital flexor (Flexor pedis perforans)	3 heads: Humeral (2, medial epicondyle) + Radial (deepest/smallest, volar radius) + Ulnar (most superficial, olecranon)	Volar surfaces, 3rd phalanges (both digits)	Flex carpus + digits, extend elbow	Median + ulnar nerves; common interosseous artery

- Deep digital flexor tendon passes **THROUGH** the rings formed by the superficial flexor tendon at the fetlock — "perforans" (deep, passes through) vs "perforatus" (superficial, gets perforated/passed-through)
- Lumbricales: absent in horse (unless counting bundles arising from deep flexor, inserting into superficial flexor tendon at carpus)
- Interossei: only **ONE** — the interosseous medius = suspensory ligament

EXAM TIP

"Perforans" (deep flexor) **PASSES THROUGH** the ring; "Perforatus" (superficial flexor) **IS** the one **PERFORATED**/passed-through. The Latin names literally describe their relationship at the fetlock.

PYQ

The deep digital flexor is also known by which name reflecting its relationship to the superficial flexor at the fetlock?

A) Flexor pedis perforatus **B) Flexor pedis perforans ✓** C) Flexor metacarpi externus D) Flexor carpi ulnaris

Exp: "Perforans" (deep digital flexor) means it passes through the rings formed by the superficial flexor (perforatus) tendon.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the deep digital flexor muscle with its three heads of origin.

Quick Recap

- Superficial layer: Pronator teres (vestigial!), Flexor carpi radialis, Flexor carpi ulnaris, Ulnaris lateralis — ALL flex carpus + extend elbow
- Deep layer: Superficial digital flexor + Deep digital flexor — flex digits
- Interosseous medius = the suspensory ligament (only 1 interosseus muscle here)

MYOLOGY — ARM, FOREARM & MANUS

Pectoral Limb Muscles — Horse Vs Dog Variations

Horse — Key Differences from Ox

Muscle	Horse Variation
Anterior superficial pectoral	More distinct than posterior
Deep pectoral	Divides into anterior (extends to scapula's cervical angle, in front of supraspinatus) + posterior (fan-shaped)
Trapezius	Cervical + dorsal parts clearly separated by an aponeurosis
Omotransversarius	ABSENT
Brachiocephalicus	Less distinct division into 2 parts
Serratus thoracis + cervicis	Fuse into ONE wide fan-shaped serratus ventralis (inserted to 2 triangular scapular areas)
Subscapularis	Only ONE fleshy part (not 3 as in ox)

Muscle	Horse Variation
Capsularis	NEW muscle (absent in ox) — rim of glenoid cavity to humeral neck; raises shoulder joint capsule during flexion
Deltoides (acromial part equivalent)	Arises from scapular spine via strong aponeurosis covering infraspinatus
Tensor fascia antibrachii	Very wide below; inserts to deep forearm fascia + olecranon
Medial digital extensor	ABSENT
Extensor pedis (=common digital extensor)	Arises with lateral digital extensor; forms broad ligament with suspensory ligament slips to extensor process of pedis
Pronator teres	Usually absent or just a tendinous band
Superficial digital flexor	Does NOT divide into 2 parts; reinforced above carpus by the superior check ligament
Deep digital flexor	Receives the inferior check ligament half-way down metacarpus; inserts to semilunar crest + os pedis tendinous surface

IMPORTANT

The horse's superior + inferior check ligaments (reinforcing the superficial and deep digital flexors respectively) are THE anatomical basis for the horse's ability to stand/rest/sleep with minimal muscular effort — a cornerstone applied-physiology fact.

Dog — Key Differences from Ox

Muscle	Dog Variation
Posterior superficial pectoral	Thin and narrow
Anterior deep pectoral	Feeble, terminates with posterior deep pectoral on humerus
Trapezius	Thin; cervical + dorsal portions meet on a fibrous raphe
Omotransversarius	Well-developed, uniform thickness (opposite of horse)
Serratus ventralis	Highly developed; NO visible separation between cervicis/thoracis parts; semicircular, C3-C7 + first 8 ribs to scapula's cervical + thoracic angles
Rhomboideus	Very extensive
Teres major	Thicker
Capsularis	ABSENT (opposite of horse)
Deltoides	Well developed; coracobrachialis very small
Biceps brachii	Very prominent
Triceps	Extra ACCESSORY head (below humeral head); heads marked by fat deposition
Brachioradialis	NEW muscle — lateral humeral epicondyle to medial radial border
Common digital extensor	Single muscle, 4 tendons to 3rd phalanges of digits 2-5
Extensor of 1st/2nd digits	Small muscle between common digital + lateral digital extensor

Muscle	Dog Variation
Lateral digital extensor	Double belly (uncommon) — 2 separate extensor bundles for digits 3-4 and digit 5
Pronator teres	Functional here (helps in pronation) — NOT vestigial like in horse
Flexor carpi ulnaris	2 heads (humeral + ulnar) to accessory carpal
Superficial digital flexor	Single origin (medial humeral condyle); reinforced by accessory carpal + proximal sesamoid slips; 4 bands to 2nd phalanges of digits 2-5
Deep digital flexor	2 attachments: 1st digit + 4 branches to digits 2-5's 3rd phalanges
Palmaris longus	NEW muscle — joins deep + superficial flexors, from mid-radius to behind carpus

EXAM TIP

Capsularis = present in HORSE, absent in dog and ox. Omotransversarius = absent in HORSE, well-developed in dog and present in ox. These two "opposite presence" facts are a classic comparative bit-question pair.

PYQ

The capsularis muscle, which raises the shoulder joint capsule during flexion, is found in which species?

A) Ox only **B) Horse only ✓** C) Dog only D) All three equally

Exp: The capsularis is a horse-specific muscle, absent in both ox and dog.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Compare the pectoral limb musculature of the horse and dog, highlighting muscles unique to each.

Quick Recap

- Horse-unique: Capsularis, Superior + Inferior check ligaments (standing rest mechanism)
- Horse-absent: Omotransversarius, Medial digital extensor
- Dog-unique: Brachioradialis, Palmaris longus, functional Pronator teres
- Dog-absent: Capsularis

MYOLOGY — PELVIC LIMB: HIP & THIGH

Fascia of the Thigh & Lateral Muscles of Hip/thigh

Fascia of the Thigh

- Superficial fascia: no special description needed
- Deep fascia — internal portion covers the first layer of medial thigh muscles; continuous below with leg fascia, continuous in front with the fascia covering anterior/lateral thigh
- **Tensor fasciae latae** = the portion of deep fascia covering lateral + anterior thigh surfaces; continuous with gluteal fascia above
- Tensor fasciae latae + one termination of biceps femoris insert into the fascia latae, which attaches to the patella + its medial/lateral ligaments
- Fascia latae becomes continuous with the fascia covering medial thigh muscles

Lateral Muscles of Hip & Thigh — Overview

- List: Tensor fasciae latae (Tensor vaginae femoris) | Biceps femoris | Gluteus medius (Middle gluteus) | Gluteus profundus (Deep gluteus) | Semitendinosus

Muscle	Origin	Insertion	Action
Tensor fasciae latae	Lateral angle of ilium	Fascia latae + lateral patellar ligament	Tense fascia latae, flex hip, extend stifle
Biceps femoris	Sacral spines, sacro-sciatic ligament, tuber ischii, gluteal fascia	3 aponeurotic tendons: patella, tibial crest, fascia latae	Extend + abduct limb (kicking/propulsion). Anterior part=extends stifle; Middle=extends hip; Posterior=tenses fascia latae
Gluteus medius (Middle gluteus)	Lateral ilium angle, sacro-sciatic ligament, gluteal ilium surface, gluteal fascia	3 branches: Anterior (tendon, below trochanter major = "gluteus accessorius"), Middle (strong tendon, trochanter major summit), Posterior (fleshy, trochanteric ridge)	Extend hip, abduct limb (kicking/propulsion)
Gluteus profundus (Deep gluteus)	Lateral ilium angle + gluteal surface in front of gluteal line; lateral ischiatic spine surface	Tubercle below trochanter major base	Abduct limb, rotate it inward
Semitendinosus	Ventral surface of tuber ischii	Tibial crest + fascia latae	Extend hip, assist kicking/propulsion

- Note (Tensor fasciae latae): the SUPERFICIAL gluteus muscle is ABSENT in ox — its anterior part fuses with tensor fasciae latae, posterior part fuses with biceps femoris
- Gluteus medius: the anterior branch insertion (below trochanter major) is separable/deep, called the **gluteus accessorius**
- Gluteus medius + Gluteus profundus blood/nerve: Anterior gluteal artery + Anterior gluteal nerve
- Biceps femoris blood/nerve: Posterior gluteal, deep femoral, posterior femoral arteries + Posterior gluteal + sciatic nerves
- Semitendinosus blood/nerve: Deep femoral artery + Sciatic nerve

IMPORTANT

Ox lacks a distinct superficial gluteus muscle — it has fused into tensor fasciae latae (anterior part) and biceps femoris (posterior part). This is a key species-specific absence.

EXAM TIP

Biceps femoris has 3 functional parts doing 3 different jobs (extend stifle / extend hip / tense fascia latae) from ONE muscle — a good example of a single muscle with regionally divided actions.

PYQ

In the ox, which muscle is absent, with its anterior part fusing into tensor fasciae latae and posterior part into biceps femoris?

- A) Gluteus medius B) Gluteus profundus **C) Superficial gluteus ✓** D) Semitendinosus

Exp: The superficial gluteus muscle is absent in ox, having fused with tensor fasciae latae (anterior) and biceps femoris (posterior).

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the biceps femoris muscle and its three functional parts.

Quick Recap

- Tensor fasciae latae = deep fascia over lateral/anterior thigh; flexes hip, extends stifle
- Biceps femoris = 3-part action (stifle/hip/fascia); Gluteus medius + profundus = hip extension/abduction/rotation
- Ox has NO distinct superficial gluteus (fused into 2 other muscles)

MYOLOGY — PELVIC LIMB: HIP & THIGH

Medial Muscles of Thigh — First & Second Layer

Overview — 3 Layers, 8 Muscles Total (Crural Region)

Layer	Muscles
First	Sartorius, Gracilis
Second	Pectineus, Adductor, Semimembranosus
Third	Quadratus femoris, Obturator internus, Obturator externus, Gemellus

- Obturator internus is placed WITHIN the pelvic cavity (unique among these)

First Layer

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Sartorius	2 heads: iliac fascia + tendon of psoas minor (femoral vessels pass between them)	Medial straight patellar ligament + fascia latae + tibial tuberosity	Flex hind limb, adduct limb	Femoral artery	Saphenous nerve
Gracilis	Pelvic symphysis + prepubic tendon	Medial patellar ligament + medial tibial surface + fascia latae	Abduct limb, tense leg fascia	Deep femoral artery	Obturator nerve

- Sartorius: long, thin, flat; narrow at distal end; proximal end in abdominal cavity at pelvic brim; extends obliquely down-back, sublumbar region to medial thigh
- Gracilis: extensively united to its fellow at origin; LARGER + WIDER of the 2 first-layer muscles
- **Femoral triangle**: space between gracilis and sartorius (proximal medial thigh) — lodges the femoral vessels

IMPORTANT

The femoral triangle (between gracilis and sartorius) is a key clinical landmark for locating the femoral vessels — relevant for injections, catheterization, and vascular access.

Second Layer

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Pectineus	Ventral pubis surface + prepubic tendon	Middle, medial border posterior femoral surface + (with sartorius/gracilis) medial tibial tuberosity	Abduct limb, flex hip, rotate femur inward	Deep femoral artery	Obturator nerve

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Adductor	Ventral surface, pubis + ischium	Lateral part, posterior femoral surface	Adduct limb, extend hip, rotate femur inward	Deep femoral artery	Obturator nerve
Semimembranosus	Ventral face of ischium + tuber ischii	Medial femoral epicondyle + medial tibial condyle	Extend hip, adduct limb	Deep femoral artery	Sciatic nerve

- Pectineus: large, thick, conical; under sartorius + gracilis
- Adductor: lies behind vastus internus + pectineus
- Semimembranosus: VERY large quadrilateral; posterior-most muscle of 2nd layer; large medial-face extent visible after removing 1st layer

Femoral Canal

- Muscular canal: bounded in front by sartorius, behind by pectineus, medially by gracilis + iliopsoas, laterally by vastus medialis
- Lower opening: at pectineus fibre insertion | Upper opening = **femoral ring** (opens into abdomen)
- Femoral ring bounded: front by Poupart's/inguinal ligament, behind by pubis anterior border, laterally by psoas minor tendon
- Lodges: femoral vessels, saphenous artery/vein/nerve
- Terminates below at the lower part of pectineus insertion

EXAM TIP

Pectineus, Adductor, and Semimembranosus ALL share the same nerve — Obturator (pectineus, adductor) except Semimembranosus which uses Sciatic — worth double-checking this exception when memorizing.

PYQ

The femoral ring, the upper opening of the femoral canal, opens into which cavity?

A) Thoracic B) Pelvic **C) Abdominal ✓** D) Scrotal

Exp: The femoral ring is the upper opening of the femoral canal and opens into the abdominal cavity.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the femoral canal and femoral ring with their boundaries and contents.

Quick Recap

- First layer: Sartorius (flex+adduct) + Gracilis (abduct) — femoral triangle between them
- Second layer: Pectineus (abduct+flex hip), Adductor (adduct+extend hip), Semimembranosus (extend hip+adduct) — LARGEST is semimembranosus
- Femoral canal: sartorius(front)-pectineus(behind)-gracilis/iliopsoas(medial)-vastus medialis(lateral); upper opening=femoral ring (abdomen)

MYOLOGY — PELVIC LIMB: HIP & THIGH

Medial Muscles of Thigh — Third Layer

Overview

- Third layer muscles lie more on the POSTERIOR aspect of the hip joint

- List: Quadratus femoris, Obturator externus, Obturator internus, Gemellus
- Obturator internus is UNIQUE — placed within the pelvic cavity (not purely thigh-region like the others)

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Quadratus femoris	Ventral face of ischium	Trochanter minor of femur	Extend hip, adduct thigh	Deep femoral artery	Sciatic nerve
Obturator externus	Ventral ischium + pubis surface, around obturator foramen	Common with obturator internus, trochanteric fossa	Adduct thigh, rotate it outward	Deep femoral artery	Obturator nerve
Obturator internus	Pelvic surface, ischium + pubis, around obturator foramen	Tendon through obturator foramen, common with externus, trochanteric fossa	Rotate femur outward	Deep femoral artery	Obturator nerve
Gemellus	Lateral ischium border (lesser sacro-sciatic foramen) + lateral posterior sacro-sciatic ligament	Trochanteric fossa	Rotate femur outward	Deep femoral artery	Sciatic nerve

- Quadratus femoris: narrow, short; under cover of adductor + semimembranosus origins
- Obturator externus: coarse muscular bundles around the obturator foramen (ventral/outer ischium surface)
- Obturator internus: extensive; floor of pelvic cavity, above/mostly behind obturator foramen — the ONLY third-layer muscle inside the pelvis
- Gemellus: small, thin

IMPORTANT

Obturator externus AND obturator internus share a COMMON insertion point (trochanteric fossa) despite obturator internus being inside the pelvis and externus being outside — their tendons converge at the same femoral landmark.

EXAM TIP

Three of the four third-layer muscles (Obturator externus, Obturator internus, Gemellus) all ROTATE the femur OUTWARD — only Quadratus femoris has a different primary action (extend hip + adduct).

PYQ

Which third-layer thigh muscle is uniquely located within the pelvic cavity?

A) Quadratus femoris B) Obturator externus **C) Obturator internus ✓** D) Gemellus

Exp: Obturator internus is situated on the floor of the pelvic cavity, unlike the other third-layer muscles which lie in the thigh region proper.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the obturator externus and internus muscles, noting their common insertion.

Quick Recap

- Third layer = posterior hip muscles: Quadratus femoris, Obturator externus, Obturator internus, Gemellus
- Obturator internus = only one INSIDE the pelvic cavity

- 3 of 4 muscles (both obturators + gemellus) = rotate femur outward; Quadratus femoris = extend hip + adduct

MYOLOGY — PELVIC LIMB: HIP & THIGH

Anterior Muscles of the Thigh

Overview

- List: Sartorius (already covered in medial group) | Quadriceps femoris (4 divisions: Rectus femoris, Vastus lateralis, Vastus medialis, Vastus intermedius) | Articularis genu

Quadriceps Femoris — 4 Divisions

- Massive muscle around anterior/lateral/medial femoral faces
- Rectus femoris arises from the ILIUM; the other 3 (vastus lateralis/medialis/intermedius) arise from the FEMUR
- ALL FOUR terminate on the patella

Division	Origin	Insertion	Action	Note
Rectus femoris	2 heads, 2 depressions at cotyloid border of ilium (above/in front of acetabulum)	Base + anterior surface of patella	Extend stifle, flex hip	ONLY division that also flexes the hip (crosses 2 joints, like triceps' long head)
Vastus lateralis	Lateral femoral border/surface, greater trochanter to supracondyloid fossa	Lateral patella surface + rectus femoris tendon	Extend stifle	THICKER + LARGER than vastus medialis
Vastus medialis	Medial + narrow anterior femoral part	Medial patellar ligament + rectus femoris tendon	Extend stifle	THINNER + NARROWER than vastus lateralis
Vastus intermedius	Anterior femoral surface	Patella base + femoro-patellar capsule	Extend stifle, raise capsule during extension	2 parts, lies DEEPLY, covered by the other 3 divisions

- Vastus intermedius blood/nerve: Anterior femoral artery + Femoral nerve
- Note: the 3 straight patellar ligaments lie in front of the quadriceps tendons of insertion; the patella develops AS a sesamoid bone within these tendons

IMPORTANT

Rectus femoris is the ONLY quadriceps division that crosses BOTH the hip and stifle joints (origin=ilium, not femur) — giving it a dual action (flex hip + extend stifle), just like triceps' long head crossing shoulder+elbow.

Articularis Genu

Feature	Detail
Location	Lower 1/3 of anterior femoral surface, under cover of vastus intermedius
Origin	Lower part, anterior femoral face
Insertion	Femoro-patellar capsule
Action	Raise the capsule
Blood supply	Anterior femoral artery
Nerve supply	Femoral nerve

- Very small muscle; functionally assists vastus intermedius (same capsule-raising action)

PYQ

Which division of the quadriceps femoris uniquely arises from the ilium rather than the femur?

A) Vastus lateralis B) Vastus medialis C) Vastus intermedius **D) Rectus femoris ✓**

Exp: Rectus femoris arises from the ilium (cotyloid border), unlike the three vastus divisions which arise from the femur — this is why it alone also flexes the hip.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the four divisions of the quadriceps femoris with their origins and actions.

Quick Recap

- Quadriceps femoris = Rectus femoris (ilium origin, flexes hip too) + Vastus lateralis/medialis/intermedius (femur origin, extend stifle only)
- ALL 4 divisions insert on the patella (which is itself a sesamoid bone within their tendons)
- Articularis genu = tiny muscle, same action as vastus intermedius (raises joint capsule)

MYOLOGY — HIP & THIGH: HORSE/DOG VARIATIONS

Pelvic Limb Hip/thigh Muscles — Horse Vs Dog Variations

Horse — Key Differences from Ox

Muscle	Horse Variation
Tensor fasciae latae	LESS extensive than in ox
Superficial gluteus	Present (unlike ox where it's absent/fused!) — distinct, 'V'-shaped; arises from ilium's lateral angle + gluteal fascia; inserts to the THIRD trochanter
Gluteus medius	Extends far forward over longissimus dorsi muscle; inserts to trochanter major; assists in BACK KICK
Gluteus profundus	Smaller; inserts to the convexity's inner face
Biceps femoris	Extra attachment to femur (rough surface near trochanter tertius); also assists in back kick
Semitendinosus	Arises from transverse process of 1st 2-3 COCCYGEAL vertebrae (not just tuber ischii)
Sartorius	Does NOT arise from the shaft of the ilium
Gracilis	NOT as extensively fused to its fellow at origin (unlike ox)
Semimembranosus	LARGE; also arises from the sacro-sciatic ligament
Obturator internus	Has iliac + sacral parts = called " piriformis "; tendon passes through LESSER sciatic foramen, does NOT unite with obturator externus (unlike ox where they share insertion)
Articularis genu	May NOT be seen as a separate muscle
Capsularis	NEW muscle (absent in ox) — arises from ilium above rectus femoris's outer part origin; tendon passes between vastus lateralis + intermedius; inserts to upper 1/3 anterior femoral face + hip capsular ligament; raises the capsule during hip flexion

IMPORTANT

Horse uniquely HAS a distinct superficial gluteus (3rd trochanter insertion) — the exact OPPOSITE of ox, where it's absent/fused into tensor fasciae latae + biceps femoris. A frequently tested contrast.

EXAM TIP

"Piriformis" is the horse-specific name for obturator internus's iliac+sacral combined part — remember this alternate name is horse-only terminology.

Dog — Key Differences from Ox

Muscle	Dog Variation
Tensor fasciae latae	Thin and long
Superficial gluteus	Present; overlies gluteus medius POSTERIORLY
Gluteus medius	Does NOT extend forward beyond the ilium — hence NO lumbar part (unlike horse!); inserts into trochanter major
Gluteus profundus	Intimately attached to middle gluteus; SMALLER of the two, adherent to ilium
Biceps femoris	Origin from tuber ischii, placed POSTERIORLY; has an extra thin band from sacro-sciatic ligament blending with it = abductor cruris posterior
Semimembranosus	CLEARLY divisible into 2 parts: anterior (ends under pectineus attachment) + posterior (attached to abductor's posterior border, terminates on femur's medial condyle)
Semitendinosus	Arises from tuber ischii; inserts on tibial fascia (medially)
Sartorius	2 parts DIVERGE after their origin
Gracilis	Thinner and narrower
Adductor	Divided into Adductor minor + Adductor major — both insert to the inner branch of the femur's linea aspera
Pectineus, Obturator, Quadratus femoris, Gemellus, Rectus femoris	Similar to ox (no major differences)

EXAM TIP

Dog's Gluteus medius has NO lumbar extension — directly opposite the HORSE's gluteus medius, which extends far forward over the longissimus dorsi. A clean species contrast pair.

PYQ

The alternate name for the obturator internus in the horse, reflecting its combined iliac and sacral parts, is?

A) Gemellus B) **Piriformis** ✓ C) Capsularis D) Quadratus femoris

Exp: In the horse, the obturator internus's iliac + sacral parts together are called the piriformis.

PYQ

Which species has a superficial gluteus muscle that is distinct and inserts onto the third trochanter?

A) Ox B) Horse C) Dog (posteriorly overlying) D) **Both B and C** ✓

Exp: Both horse (distinct, 3rd trochanter insertion) and dog (overlying gluteus medius posteriorly) have a superficial gluteus; ox uniquely lacks a distinct one.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Compare the gluteus medius muscle across ox, horse, and dog.

Quick Recap

- Superficial gluteus: ABSENT in ox (fused) | PRESENT in horse (3rd trochanter) | PRESENT in dog (overlies gluteus medius)
- Gluteus medius: extends over longissimus dorsi in HORSE (assists back kick) | NO lumbar part in DOG
- Capsularis: horse-ONLY muscle (raises hip capsule on flexion)
- Obturator internus = "piriformis" in horse (iliac+sacral parts, doesn't unite with externus)
- Dog's Adductor uniquely splits into minor + major

MYOLOGY — PELVIC LIMB: LEG & FOOT

Fascia of the Leg & Dorsolateral Group Overview

Fascia of the Leg

- Several superimposed layers; superficial fascia continuous with thigh's
- 2nd layer = tendons of tensor fasciae latae, biceps femoris, semitendinosus, gracilis, sartorius (attach to patellar ligaments, tibial crest, medial tibial face)
- Both layers unite behind the deep digital flexor at about mid-region
- 3rd layer = furnishes muscle sheaths
- Tarsal fascia continues down over metatarsals, blends with ligaments, attaches to bone eminences — thick front+back, thin laterally
- Metatarsal + digital fascia = similar to forelimb's

Two Muscle Groups of the Leg

Group	Muscles
Dorsolateral	Complex muscle (peroneus tertius + long digital extensor + medial digital extensor, all one origin), Tibialis anterior, Peroneus longus, Lateral digital extensor, Extensor digitalis brevis
Plantar	Superficial layer: Gastrocnemius, Soleus, Superficial digital flexor. Deep layer: Popliteus, Deep digital flexor

- These muscles almost completely envelop the tibia, leaving only the medial face uncovered

EXAM TIP

"Dorsolateral" muscles = mostly EXTENSORS (flex hock/extend digits); "Plantar" muscles = mostly FLEXORS (extend hock/flex digits) — opposite functional groups, mirroring the forearm's extensor/flexor split.

Quick Recap

- Leg fascia: 3 layers, unite behind deep digital flexor mid-region
- 2 muscle groups: Dorsolateral (dorso-lateral tibia) + Plantar (behind tibia, 2 layers)
- Only the MEDIAL tibial face stays uncovered by muscle

MYOLOGY — PELVIC LIMB: LEG & FOOT

Dorsolateral Group — Complex Muscle & Tibialis Anterior

Complex Muscle — 3 Bellies from ONE Origin

- Single muscle, dorso-lateral leg aspect; origin = strong tendon from femur's extensor fossa
- Splits into 3 muscular bellies mid-way, each succeeded by its own tendon

Belly	Position	Insertion	Note
Peroneus tertius	Most superficial, well-developed; front + partly internal to other 2	Tubercle, dorso-medial proximal large metatarsal	Tendon perforated BY tibialis anterior; synovial sheath at tarsus
Long digital extensor (Extensor digitalis longus)	Lateral to peroneus tertius	Extensor process, 3rd phalanx	Tendon behaves like forelimb's common digital extensor
Medial digital extensor	Under cover of both preceding divisions	Resembles forelimb's course/termination	Shares common synovial sheath (at tarsus) with long digital extensor

- Action (whole complex muscle): Flex the HOCK, extend the DIGITS
- Blood supply: Anterior tibial artery | Nerve supply: Deep peroneal nerve

Tibialis Anterior (Tibialis Anticus)

Feature	Detail
Location	Thin, triangular; under complex muscle, lateral tibial surface; 2 heads of origin
Origin	Tibial crest tuberosity + upper lateral tibial border + fibrous cord of fibula (tendon passes THROUGH peroneus tertius's tendon)
Insertion	Postero-medial surface, upper large metatarsal + 2nd/3rd tarsal bones
Action	Flex the hock
Blood supply	Anterior tibial artery
Nerve supply	Deep peroneal nerve

IMPORTANT

Tibialis anterior's tendon literally passes THROUGH (perforates) the peroneus tertius tendon on its way down — an unusual anatomical arrangement worth remembering as a specific mechanical relationship, not just a location fact.

PYQ

The action of the complex muscle (peroneus tertius + long digital extensor + medial digital extensor together) is to?

- A) Extend hock, flex digits **B) Flex hock, extend digits ✓** C) Flex hock, flex digits D) Extend hock, extend digits

Exp: The complex muscle flexes the hock and extends the digits.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the three bellies of the complex muscle and their common origin.

Quick Recap

- Complex muscle = 1 origin, 3 bellies (peroneus tertius, long digital extensor, medial digital extensor); flexes hock, extends digits
- Tibialis anterior: flexes hock; tendon perforates peroneus tertius's tendon

Dorsolateral Group — Peroneus Longus, Lateral Digital Extensor & Short Extensor

Muscle	Location	Origin	Insertion	Action	Blood/Nerve
Peroneus longus	Between peroneus tertius + lateral digital extensor	Lateral tibial condyle + fibrous cord of fibula	1st tarsal + proximal large metatarsal	Rotate the hock INWARD	Anterior tibial artery; Deep peroneal nerve
Lateral digital extensor (Extensor digitalis lateralis)	Lateral to peroneus longus	Lateral stifle ligaments + lateral tibial condyle	Resembles forelimb muscle's course/termination	(extends lateral digit, per forelimb pattern)	Anterior tibial artery; Superficial peroneal nerve
Extensor digitalis brevis (Short extensor)	Narrow, dorsal metatarsal surface	Depression, dorsal metatarsal surface	Tendon of long digital extensor	Assists the long digital extensor	Anterior metatarsal artery; Deep peroneal nerve

- Peroneus longus tendon: crosses OVER lateral digital extensor's tendon, passes under the lateral ligament, THROUGH a bony groove (between tarsal bones + large metatarsal)

EXAM TIP

Peroneus longus rotates the hock INWARD — the only rotational action named explicitly among the dorsolateral group, distinguishing it from the mostly flexion/extension actions of its neighbors.

PYQ

Which dorsolateral leg muscle's specific action is to rotate the hock inward?

A) Tibialis anterior **B) Peroneus longus ✓** C) Lateral digital extensor D) Extensor digitalis brevis

Exp: Peroneus longus specifically rotates the hock inward, distinct from the flex/extend actions of the other dorsolateral muscles.

Quick Recap

- Peroneus longus = rotates hock inward (tendon crosses lateral digital extensor's tendon, through a bony groove)
- Lateral digital extensor = mirrors forelimb's lateral digital extensor pattern
- Extensor digitalis brevis = small assistant to the long digital extensor

Plantar Group — Superficial Layer

Overview

- 2 superimposed layers behind the tibia: Superficial (Gastrocnemius, Soleus, Superficial digital flexor) + Deep (Popliteus, Deep digital flexor)

Muscle	Origin	Insertion	Action	Blood/Nerve
Gastrocnemius	2 heads: medial (medial supracondyloid crest) + lateral (supracondyloid fossa's outer lip)	Summit of the tuber calcis	Extend the hock	Popliteal + posterior femoral arteries; Tibial nerve
Soleus	Lateral tibial condyle + proximal fibula	Lateral gastrocnemius head, above its middle	Assist gastrocnemius in extending hock	Popliteal artery; Tibial nerve

Muscle	Origin	Insertion	Action	Blood/Nerve
Superficial digital flexor	Supracondyloid fossa (closely blended between gastrocnemius's 2 heads)	Passes over tuber calcis summit (2 slips attach there), down metatarsal region (as in forelimb)	Extend hock, flex digits	Popliteal artery; Tibial nerve

The "Tendo Achilles" Complex

- Gastrocnemius's 2 heads unite mid-leg → common tendon, placed immediately **BEHIND** the superficial flexor's tendon
- A **TWIST** occurs at the lower 1/3 of the leg: superficial flexor's tendon ends up **BEHIND** the gastrocnemius tendon
- Superficial digital flexor's tendon + its fibrous bands (attaching to os calcis sides) completely **COVER** the gastrocnemius tendon
- Together = **tendo calcaneus / hamstring / tendo Achilles**

IMPORTANT

The "tendo Achilles" in veterinary anatomy is actually a **COMBINED** structure — gastrocnemius tendon + superficial digital flexor tendon twisted together, not gastrocnemius alone as in human anatomy. A key comparative-anatomy distinction.

PYQ

The tendo Achilles (tendo calcaneus) in domestic animals is formed by the combination of which two tendons?

- A) Gastrocnemius + Deep digital flexor **B) Gastrocnemius + Superficial digital flexor ✓** C) Soleus + Popliteus D) Tibialis anterior + Peroneus longus

Exp: The tendo Achilles is formed by the twisted combination of the gastrocnemius tendon and the superficial digital flexor tendon.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the formation of the tendo Achilles (tendo calcaneus) in the ox.

Quick Recap

- Superficial plantar layer: Gastrocnemius + Soleus (both extend hock) + Superficial digital flexor (extends hock, flexes digits)
- Tendo Achilles = Gastrocnemius tendon + Superficial digital flexor tendon (twisted together, **NOT** gastrocnemius alone)

MYOLOGY — PELVIC LIMB: LEG & FOOT

Plantar Group — Deep Layer

Muscle	Origin	Insertion	Action	Blood/Nerve
Popliteus	Lower of 2 depressions, femur's lateral condyle	Triangular area, upper 1/3 posterior tibial surface (bounded by popliteal line)	Flex the STIFLE, rotate leg inward	Posterior tibial artery; Tibial nerve

Muscle	Origin	Insertion	Action	Blood/Nerve
Deep digital flexor	3 heads: Superficial (tibialis posterior), Deep (flexor hallucis), Medial (long digital flexor/flexor accessorius) — from lateral tibial border/posterior face + lateral tibial condyle + fibrous fibular cord	Tendons unite above tarsus, common tendon through tarsal sheath, descends metatarsal region (medial head's tendon passes under lateral hock ligaments, joins at upper metatarsus 1/3)	Flex the digits, extend the hock	Posterior tibial artery; Tibial nerve

- Deep digital flexor: medial head's tendon has its OWN synovial sheath at the tarsus; superficial + deep heads share a COMMON synovial sheath
- United tendon behaves the same as the forelimb's deep digital flexor

EXAM TIP

Popliteus is the ONLY leg muscle that acts on the STIFLE joint (not the hock) — a distinguishing fact since every other leg/foot muscle in this list acts on the hock or digits.

PYQ

Which deep plantar leg muscle uniquely acts on the stifle joint rather than the hock or digits?

A) Deep digital flexor **B) Popliteus ✓** C) Soleus D) Gastrocnemius

Exp: Popliteus flexes the stifle and rotates the leg inward — the only leg/foot muscle acting on the stifle rather than the hock/digits.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the three heads of origin of the deep digital flexor in the leg.

Quick Recap

- Popliteus = flexes STIFLE (unique among leg muscles), rotates leg inward
- Deep digital flexor = 3 heads (superficial/deep/medial), flexes digits + extends hock, mirrors forelimb pattern

MYOLOGY — LEG & FOOT: HORSE/DOG VARIATIONS

Leg/foot Muscles — Horse Vs Dog Variations

Horse — Key Differences from Ox

Muscle	Horse Variation
Long digital extensor	Arises WITH the next muscle (common origin); receives the tendon of lateral digital extensor + extensor brevis a little below the hock (on its lateral aspect)
Peroneus tertius (Tendo femoro-metatarsus)	ENTIRELY TENDINOUS (no muscle belly at all!) — a unique feature
Tibialis anterior	Short wide tendon of insertion (to 1st tarsal bone) = the cunean tendon
Lateral digital extensor (Peroneus)	Joins the tendon of the long digital extensor
Extensor brevis	Joins BOTH tendons of the long and lateral digital extensor
Peroneus longus	ABSENT
Posterior group muscles	Resemble those of the ox in general

IMPORTANT

The horse's peroneus tertius is ENTIRELY TENDINOUS — no muscle fibres at all, just a fibrous cord (tendo femoro-metatarsus) linking femur to metatarsus. This is a classic biomechanical fact: it works as a passive "reciprocal apparatus" component, linking stifle flexion to hock flexion automatically without muscular effort — key to the horse's standing-rest mechanism (alongside the check ligaments).

EXAM TIP

"Cunean tendon" = horse-specific name for the tibialis anterior's insertion tendon (to the 1st tarsal bone) — this exact tendon is the site of "cunean tenectomy," a real surgical procedure for certain hock lameness conditions in horses.

Dog — Key Differences from Ox

Muscle	Dog Variation
Tibialis anterior	LARGE muscle; covers the lateral surface of common digital extensor at its proximal extremity
Common digital extensor	Mostly overlapped by tibialis anterior; inserts to 3rd phalanges of digits 2, 3, 4, 5
Peroneus longus	Has an ACCESSORY muscle called peroneus brevis (unlike horse, where peroneus longus is entirely absent!)
Soleus	ABSENT
Gastrocnemius	Has a small SESAMOID BONE called the fabella embedded in it at its origin

IMPORTANT

Fabella = a small sesamoid bone embedded in the gastrocnemius origin in the DOG — this is the same "fabella" concept mentioned earlier for the dog's femur (behind the femoral condyles); it's essentially the attachment point sesamoid for this muscle, unique to the dog among these species.

EXAM TIP

Peroneus longus is ABSENT in horse but has an EXTRA accessory muscle (peroneus brevis) in dog — a sharp species contrast on the same muscle group.

PYQ

The peroneus tertius in the horse is unique in being?

- A) Absent **B) Entirely tendinous, with no muscle belly ✓** C) Fused with peroneus longus D) Doubled into two bellies

Exp: The horse's peroneus tertius (tendo femoro-metatarsus) is entirely tendinous, functioning as a passive mechanical link between the stifle and hock joints.

PYQ

The cunean tendon, a surgical landmark in equine hock lameness, belongs to which muscle?

- A) Peroneus tertius **B) Tibialis anterior ✓** C) Long digital extensor D) Peroneus longus

Exp: The cunean tendon is the short, wide insertion tendon of the tibialis anterior onto the first tarsal bone.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the peroneus tertius of the horse and explain its unique tendinous nature.

Quick Recap

- Horse: Peroneus tertius = entirely tendinous (no belly); Peroneus longus = ABSENT; Tibialis anterior's tendon = "cunean tendon"
- Dog: Peroneus longus HAS accessory peroneus brevis; Soleus = ABSENT; Gastrocnemius has the fabella (sesamoid bone)
- Peroneus longus: absent in horse, present WITH an extra accessory muscle in dog — opposite extremes

MYOLOGY — SUBLUMBAR REGION

Muscles of the Sublumbar Region

Overview

- Deep muscles: ventral face of lumbar vertebrae + anterior ilium face
- Extend forward + backward beyond the sublumbar region proper; form part of the abdominal cavity's ROOF
- In direct contact with abdominal viscera
- List: Psoas minor, Psoas major, Iliacus, Quadratus lumborum, Intertransversales lumborum

Fascia

- **Iliac fascia:** strong fibrous expansion covering iliacus + psoas major's ventral face; attaches to psoas minor's tendon (medially) + ilium's lateral angle/border (laterally)
- Thin over psoas major's anterior part + posteriorly (over psoas major/iliacus's terminal portion)
- Its inferior part furnishes ONE origin site for the sartorius muscle

Muscle	Origin	Insertion	Action	Blood Supply	Nerve Supply
Psoas minor	Intervertebral fibro-cartilage, T12-T13 + all lumbar vertebrae	Psoas tubercle, ilium shaft	Flex pelvis on the loin, incline lumbar region laterally	Lumbar arteries	Lumbar nerves
Psoas major	Upper posterior borders, last 2 ribs + lumbar vertebrae transverse processes	Common with iliacus, trochanter minor of femur	Flex the hip, rotate thigh outward	Lumbar + circumflex iliac arteries	Ilio-psoas nerve
Iliacus	Iliac surface + ilium shaft (between psoas tubercle + rectus femoris origin) + psoas minor tendon + sacral wing; begins under L6 body	Common with psoas major, trochanter minor of femur	Same as psoas major (flex hip, rotate thigh outward)	Lumbar + deep femoral arteries	Ilio-psoas nerve
Quadratus lumborum	Ventral surface, upper part of last 2-3 ribs + lumbar transverse processes	Anterior border, first 5 lumbar transverse processes + sacral wing	SINGLY: flex loin laterally. TOGETHER: fix lumbar vertebrae + last 2 ribs	Lumbar arteries	Lumbar nerves
Intertransversales lumborum	Anterior border, transverse process of the preceding lumbar vertebra	Posterior border, transverse process of the vertebra in front	Aid lateral flexion of the loin	Lumbar arteries	Lumbar nerves

- Psoas major + Iliacus TOGETHER form the functional unit called **iliopsoas** (share the same insertion + action)

- Iliacus: appears to have 2 parts forming a groove to receive psoas major + psoas minor

IMPORTANT

Psoas major and Iliacus share IDENTICAL insertion (trochanter minor) and action — together they form the "iliopsoas," a single functional unit despite being anatomically distinct muscles. A classic combined-muscle concept, same pattern as "quadriceps femoris" grouping 4 separate muscles.

EXAM TIP

Psoas MINOR acts on the PELVIS/loin (flexes pelvis on loin); Psoas MAJOR acts on the HIP joint (flexes hip). Same name family, different joint targets — don't confuse the two.

Species Differences

Species	Key Feature
Horse	Psoas minor extends from the LAST THREE dorsal (thoracic) vertebrae; Psoas major completely COVERS quadratus lumborum; Quadratus lumborum is NARROWER; Iliacus does NOT arise from L6, less closely blended with psoas major (vs ox)
Dog	Psoas minor relatively LARGER; its anterior part blends with quadratus lumborum but does NOT extend into thoracic cavity; Psoas major is SHORT (begins from L3 or L4 only); Quadratus lumborum is LARGER/STRONGER than any other species — arises from last 3-4 thoracic vertebrae WITH psoas minor, terminates on upper pelvic ilium surface; Intertransversales lumborum = same as ox

PYQ

Psoas major and iliacus share a common insertion at which structure?

- A) Psoas tubercle **B) Trochanter minor of the femur ✓** C) Trochanter major D) Ilium's lateral angle

Exp: Psoas major and iliacus both insert at the trochanter minor of the femur, together forming the iliopsoas.

PYQ

In which species is the quadratus lumborum described as larger and stronger than in any other domestic species?

- A) Ox B) Horse **C) Dog ✓** D) All equal

Exp: The dog's quadratus lumborum is notably larger/stronger, arising from the last 3-4 thoracic vertebrae along with psoas minor.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the psoas major and iliacus muscles, explaining the concept of "iliopsoas."

Quick Recap

- Sublumbar muscles: Psoas minor (loin/pelvis), Psoas major + Iliacus = "iliopsoas" (hip flexion), Quadratus lumborum (loin flexion/fixation), Intertransversales lumborum (lateral flexion)
- Horse: psoas major covers quadratus lumborum | Dog: quadratus lumborum = LARGEST/STRONGEST among species; psoas major = shortest (starts at L3/L4)

MYOLOGY — ABDOMEN

Tunica Abdominalis, Prepubic Tendon & Cremaster Externus

Tunica Abdominalis

- Yellow ELASTIC sheet — the "abdominal tunic," lies underneath the cutaneous trunk muscle
- Very thick, closely adherent to the external oblique's aponeurotic part; only loosely covers its muscular part
- Thickest near the common prepubic tendon + linea alba; thins to almost nothing over the external oblique's muscular portion, then disappears
- Anteriorly: extends over some intercostals + serratus thoracis's posterior digitations
- Posteriorly: attaches to ilium's external angle + lumbo-dorsal fascia
- Gives rise to: **suspensory ligament of the prepuce** (male) + **suspensory ligament of the mammary gland** (female)
- Function: elastic bandage, supports abdominal muscles bearing visceral weight
- Thin in young animals, increases with digestive organ distension
- Takes the PLACE of deep fascia in this region

IMPORTANT

The tunica abdominalis is the anatomical origin of BOTH the prepuce's suspensory ligament (male) and the udder's suspensory ligament (female) — directly relevant to udder support/prolapse discussions in dairy cattle.

Prepubic Tendon

- = tendon of insertion for BOTH rectus abdominis muscles; very strong, thick band
- Directed upward-backward; inserted to the anterior border of both pubic bones (between ilio-pectineal eminences)
- Receives: aponeuroses of both oblique muscles + the inguinal ligament
- Gives origin to: pectineus muscle fibres; fuses with gracilis's medial anterior origin face
- This gracilis attachment DEPRESSES the lower abdominal wall

Cremaster Externus

Feature	Detail
Description	Narrow flat muscle band, detached from the internal oblique muscle; passes into inguinal canal with the spermatic cord
Origin	Internal oblique muscle + iliac fascia
Insertion	Parietal layer of tunica vaginalis
Action	Raise the tunica vaginalis
Blood supply	External spermatic artery
Nerve supply	External spermatic nerve

PYQ

The suspensory ligaments of both the prepuce and the mammary gland originate from which structure?

A) Rectus abdominis B) **Tunica abdominalis ✓** C) Prepubic tendon D) Linea alba

Exp: Both suspensory ligaments (prepuce in males, mammary gland in females) derive from the external face of the tunica abdominalis.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the tunica abdominalis and its clinical relevance to udder support.

Quick Recap

- Tunica abdominalis = elastic "abdominal tunic"; origin of BOTH prepuce's and udder's suspensory ligaments
- Prepubic tendon = shared insertion point for both rectus abdominis muscles + both oblique aponeuroses + inguinal ligament
- Cremaster externus = detached band of internal oblique, raises the tunica vaginalis (spermatic cord related)

MYOLOGY — ABDOMEN

Inguinal Canal

Overview

- Paired, slit-like passages through the posterior abdominal floor, either side of the prepubic tendon
- Anlagen (developmental precursor) present BEFORE sex differentiation — equally developed in both sexes initially
- Term "canal" is misleading in large quadrupeds — it's a POTENTIAL space (between the two oblique abdominis muscles), not an actual open tunnel
- Rings are SLIT-LIKE, not circular

Contents by Sex

Sex	Contents
Male	Spermatic cord, tunica vaginalis, external cremaster muscle, external pudic artery, artery of the cord, veins
Female	Rudimentary; contains only the mammary vessels + nerves
Both sexes	External pudic artery, vein, inguinal nerves, lymph vessels

Walls & Two Openings

Wall/Ring	Formed By
Antero-medial wall	Posterior part of the internal oblique muscle
Postero-lateral wall	Portion of external oblique's aponeurosis = the inguinal (Poupart's) ligament — not a separate structure, just the aponeurosis portion related to this canal, lying against the medial thigh surface
Internal (abdominal) inguinal ring (upper/peritoneal opening)	Anteriorly: free borders of internal oblique. Posteriorly: upper border of inguinal ligament. Medial angle: lateral prepubic tendon margin. Lateral angle: internal oblique's attachment to inguinal ligament. Long axis: forward-outward-upward from prepubic tendon toward tuber coxae, ~15 cm long
External inguinal ring (lower opening)	Slit-like opening in external oblique's aponeurosis, ~15 cm long, directed forward-upward-slightly downward from prepubic tendon. Medial angle = rounded (formed by prepubic tendon). Lateral angle = acute, well-defined

- Both ring openings are POTENTIAL (not gaping) — abdominal viscera's weight keeps the internal oblique in contact with the inguinal ligament except where canal contents separate the walls
- Canal length between rings VARIES: at medial angle = equal to prepubic tendon thickness (LESS than 1.25 cm); at lateral angles = ~15 cm

IMPORTANT

The inguinal (Poupart's) ligament is NOT a separate anatomical structure — it's simply the portion of the external oblique's own aponeurosis that happens to relate to this canal. A commonly misunderstood point.

EXAM TIP

Internal ring = "peritoneal opening" (upper, abdominal side); External ring = the outer opening. Both are POTENTIAL spaces, not actual gaps, in large quadrupeds — the term "canal" itself is somewhat misleading here.

Species Differences

Species	Key Feature
Horse	External oblique EXTENSIVE, reaches up to lumbar transverse processes; internal oblique LESS developed; rectus abdominis arises from 4th/5th rib cartilage to the 9th; transversus abdominis meets the diaphragm's costal attachment at its origin
Dog	Tunica abdominalis PRACTICALLY ABSENT; external oblique well-developed with a clear aponeurotic-end border; rectus abdominis extends from xiphoid cartilage to join linea alba; inguinal canal (in the BITCH) lodges the round ligament of the uterus, enclosed in a peritoneal tubular process

PYQ

In the bitch, the inguinal canal characteristically contains which structure?

A) Spermatic cord B) **Round ligament of the uterus ✓** C) External cremaster muscle D) Mammary vessels only

Exp: In the bitch, the inguinal canal lodges the round ligament of the uterus, enclosed in a tubular peritoneal process.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the boundaries and contents of the inguinal canal, noting sex differences.

Quick Recap

- Inguinal canal = potential space between oblique muscles, either side of prepubic tendon; anlagen present before sex differentiation
- Internal ring (peritoneal/upper) vs External ring (lower) — both slit-like, ~15cm long at their widest
- Male contents: spermatic cord + tunica vaginalis + cremaster; Female: rudimentary, mammary vessels only (round ligament of uterus in bitch)
- Dog: tunica abdominalis practically absent

MYOLOGY — ABDOMEN

The Four Abdominal Muscles & Linea Alba

Overview

- 4 muscles per side of the linea alba, arranged in layers (outside to inside): **Obliquus abdominis externus** → **Obliquus abdominis internus** → **Rectus abdominis** → **Transversus abdominis**
- (Cremaster externus, covered separately, is a detachment of the internal oblique)

Linea Alba

- Rectus abdominis = entirely FLESHY, floor of abdomen; other 3 muscles have a muscular part (lateral) + aponeurotic part (ventral)
- Opposite-side aponeurotic parts meet → median fibrous raphe = **linea alba** ("white line")

- Extends: xiphoid cartilage → prepubic tendon (inserted to pubis's anterior border)
- Shows a **cicatrix** (the **umbilicus**) a little behind its middle — marks the fetal umbilical opening position

The Four Muscles — Detail Table

Muscle	Origin	Insertion	Action	Blood/Nerve
Obliquus abdominis externus (External oblique)	Lateral rib faces, lower 5th rib to last rib above its middle + lumbo-dorsal fascia	Linea alba, prepubic tendon, tuber coxae	Reduces abdominal cavity size, pressures viscera (micturition/defecation/forced expiration/coughing/parturition); flexes trunk (arches back); singly = lateral trunk flexion	Circumflex iliac, intercostal, lumbar arteries; last 5-6 intercostal + first 2 lumbar spinal nerves
Obliquus abdominis internus (Internal oblique)	Ilium's lateral angle + adjacent Poupart's ligament part	Linea alba, prepubic tendon, last rib's posterior border	Similar to external oblique	Circumflex iliac, intercostal, lumbar arteries; first 2 lumbar + last thoracic spinal nerves
Rectus abdominis	Ventral/lateral sternum parts, as far forward as 3rd costal cartilage	Pubis, via prepubic tendon	Similar to preceding	Anterior + posterior abdominal arteries; last 6-7 thoracic + first 2-3 lumbar nerves
Transversus abdominis	Medial faces, asternal rib cartilages + first 4 lumbar transverse processes	Linea alba	Same as preceding	Circumflex iliac, musculo-phrenic, intercostal arteries; first 2 lumbar + last 7-8 intercostal nerves

Special Notes

- External oblique's aponeurosis (at prepubic tendon insertion) detaches a fibrous expansion = the **Poupart's/inguinal ligament** — forms the posterior wall of the inguinal canal; perforated by the subcutaneous/external inguinal ring (~5cm from midline, for spermatic cord exit)
- Internal oblique's medial part forms the inguinal canal's **ANTERIOR** wall; its upper opening here = the internal inguinal ring (slit-like)
- Rectus abdominis: 2 muscles separated (except at pelvis, by 2-4 inch interval — wall entirely fibrous there); shows 5 tendinous **inscriptions** (linea transversae) on its superficial face; near the 2nd inscription = a foramen for the subcutaneous abdominal vein

IMPORTANT

Rectus abdominis is the **ONLY** one of the 4 abdominal muscles that is entirely fleshy with no aponeurotic part — the other 3 all have a muscular (lateral) + aponeurotic (ventral) portion.

EXAM TIP

All 4 abdominal muscles share essentially the **SAME** functional action (reduce abdominal cavity, pressure viscera, flex trunk) — the differentiation between them is mainly about **LOCATION/layer** and fibre direction, not distinct actions.

PYQ

The umbilicus, marking the position of the fetal umbilical opening, is found on which structure?

- A) Prepubic tendon **B) Linea alba ✓** C) Tunica abdominalis D) Inguinal ligament

Exp: The linea alba shows the umbilicus (a cicatrix) a little behind its middle, marking the fetal umbilical opening's former position.

PYQ

Which of the four abdominal muscles is entirely fleshy, without an aponeurotic portion?

A) External oblique B) Internal oblique **C) Rectus abdominis ✓** D) Transversus abdominis

Exp: Rectus abdominis is entirely fleshy; the other three muscles all have both muscular and aponeurotic portions.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the four abdominal muscles and explain the formation of the linea alba.

Quick Recap

- 4 muscles, outside to inside: External oblique → Internal oblique → Rectus abdominis → Transversus abdominis
- Linea alba = fusion of all 4 muscles' aponeuroses; shows the umbilicus (fetal opening marker)
- Rectus abdominis = only entirely FLESHY one (others have aponeurotic parts too)
- Poupart's/inguinal ligament = external oblique's own aponeurosis extension (forms inguinal canal's posterior wall)

MYOLOGY — BACK, LOIN & TAIL

Back & Loin Muscles — Five Layers Overview + Layers 1-3

Five Superimposed Layers (Either Side of Spine)

Layer	Muscles
First	Trapezius thoracis, Latissimus dorsi
Second	Rhomboideus thoracis
Third	Serratus dorsalis anterior, Serratus dorsalis posterior
Fourth	Longissimus costarum, Longissimus dorsi
Fifth	Levatores costarum, Multifidus dorsi, Intertransversales lumborum

Layers 1 & 2 — Cross-Reference

- **Trapezius thoracis + Rhomboideus thoracis:** already fully described under Muscles of the Pectoral Girdle (same muscles — the "thoracis" part named here is simply their dorsal continuation)
- **Latissimus dorsi:** already fully described under Muscles of the Pectoral Girdle — briefly, extends lateral thorax wall, dorso-lumbar spine to arm; origin = lumbodorsal fascia + ribs 11-12 + intercostal/external oblique fascia; action = flex the shoulder joint

EXAM TIP

Don't re-memorize Trapezius/Latissimus dorsi/Rhomboideus as "new" muscles here — they're the SAME muscles from the shoulder girdle module, just being placed in the back's layer system for completeness.

Layer 3 — Serratus Dorsalis Anterior & Posterior (NEW muscles)

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Serratus dorsalis anterior	Thin, quadrilateral, flat; under rhomboideus thoracis, serratus thoracis, latissimus dorsi; aponeurotic + fleshy portions	Lumbo-dorsal fascia	3 fleshy digitations, anterior borders + lateral surface, ribs 5-7	INSPIRATORY muscle	5th-7th intercostal arteries; 5th-7th dorsal spinal nerves
Serratus dorsalis posterior	Thin, elongated, quadrilateral — WIDER than the anterior one; aponeurotic + fleshy portions	Lumbo-dorsal fascia	Posterior borders + lateral surface, last 3 ribs	EXPIRATORY muscle	10th-13th intercostal arteries; 10th-13th dorsal spinal nerves

IMPORTANT

Serratus dorsalis anterior = INSPIRATORY; Serratus dorsalis posterior = EXPIRATORY — opposite respiratory roles despite similar names/structure. A frequently tested contrast pair.

Species Differences (Layer 3)

Species	Key Feature
Horse	Serratus dorsalis anterior MORE extensive, 8 digitations; Serratus dorsalis posterior has 7-8 digitations, its anterior ones overlap the anterior serratus dorsalis; Levatores costarum = 17 pairs
Dog	Serratus dorsalis anterior arises from first 6-7 dorsal spines, inserted ribs 2-5, VERY THICK/well-developed; Serratus dorsalis posterior arises from lumbo-dorsal fascia, inserted to last 3-4 ribs (3-4 digitations); Levatores costarum similar to ox/horse but lumbar part NOT depressed for gluteus medius lodgement; other muscles similar to ox

PYQ

Which layer-3 back muscle acts as an inspiratory muscle?

A) Serratus dorsalis posterior **B) Serratus dorsalis anterior ✓** C) Both equally D) Neither

Exp: Serratus dorsalis anterior is inspiratory; serratus dorsalis posterior is expiratory.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the serratus dorsalis anterior and posterior muscles with their opposing respiratory actions.

Quick Recap

- 5 layers of back/loin muscles; Layers 1-2 (Trapezius thoracis, Latissimus dorsi, Rhomboideus thoracis) = already covered in pectoral girdle module
- Layer 3 = Serratus dorsalis anterior (INSPIRATORY) + posterior (EXPIRATORY) — opposite actions
- Horse: 17 pairs levatores costarum | Dog: very thick serratus dorsalis anterior

MYOLOGY — BACK, LOIN & TAIL

Back & Loin Muscles — Layers 4 & 5

Layer 4 — Longissimus Costarum & Longissimus Dorsi

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Longissimus costarum	Long, narrow, thin; along longissimus dorsi's lateral border, across the ribs; distinct LUMBAR + COSTAL parts	Lumbar part: lumbar transverse processes + ilium's lateral angle. Costal part: anterior borders + lateral surfaces, last 6-7 ribs	Costal part: posterior border, ribs 1-10 + last cervical transverse process. Lumbar part: posterior border of last rib	EXPIRATORY muscle	Lumbar + intercostal arteries; Lumbar + dorsal spinal nerves
Longissimus dorsi	LONGEST + MOST POWERFUL muscle in the body! Sacrum/ilium to mid-cervical spine; medial to longissimus costarum, not distinctly separable from it in lumbar region	Ilium's crest + lateral angle, sacral spine summits, lumbo-dorsal spines, supraspinous ligament	Transverse + articular processes (lumbar vertebrae), transverse processes (dorsal vertebrae), rib upper surfaces. At ~rib 7 level, divides into dorsal part (Spinalis et semispinalis) + ventral part (longissimus cervicis), with the COMPLEXUS muscle emerging between them	BOTH sides: extend back/loin. SINGLY: flex spine to same side. Attachment to ribs = also acts as EXPIRATORY muscle	Dorsal, deep cervical, vertebral, intercostal, lumbar arteries; Cervical, dorsal, lumbar spinal nerves

- Longissimus dorsi — 3 fasciculi sets: **Superficial/spinal** (attached to spinous process), **Deep/transverse** (attached to articular + transverse processes, lumbar/thoracic/cervical), **Costal** (turn outward, attach to rib lateral surfaces)

IMPORTANT

Longissimus dorsi is explicitly named as the LONGEST and MOST POWERFUL muscle in the entire body — a striking superlative fact worth remembering verbatim for bit questions.

Layer 5 — Levatores Costarum, Multifidus Dorsi & Intertransversales Lumborum

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Levatores costarum	10-11 pairs, small/flat/triangular musculo-tendinous fasciculi; upper intercostal space ends; blend with external intercostals at insertion	Transverse process, thoracic vertebrae	Anterior border of the succeeding rib	Aid INSPIRATION	Intercostal arteries; Intercostal nerves

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Multifidus dorsi	Long muscle, directly on vertebral spinous processes, sacrum to neck; multiple forward/upward-directed segments/bundles; continuous with multifidus cervicis (front) + sacro-coccygeus lateralis (behind)	Lateral sacrum part, lumbar vertebrae's oblique process, dorsal vertebrae transverse processes	Posterior border, neural spines of lumbar + thoracic vertebrae in front	Extend the back, flex it to a side	Intercostal + lumbar arteries; Dorsal + lumbar spinal nerves
Intertransversales lumborum	(Already covered under Sublumbar Muscles — cross-reference) Small flat muscles filling inter-transverse spaces of lumbar vertebrae	Anterior border, preceding lumbar vertebra's transverse process	Posterior border, transverse process of vertebra in front	Aid lateral flexion of loin	Lumbar arteries; Lumbar nerves

EXAM TIP

Multifidus dorsi is CONTINUOUS front-to-back with other named muscles — multifidus cervicis anteriorly, sacro-coccygeus lateralis posteriorly — forming one long functional chain from neck to tail, even though each named segment is studied separately.

PYQ

Which muscle is described as the longest and most powerful muscle in the body?

- A) Longissimus costarum **B) Longissimus dorsi ✓** C) Multifidus dorsi D) Latissimus dorsi

Exp: Longissimus dorsi, extending from the sacrum/ilium to the mid-cervical spine, is the longest and most powerful muscle in the body.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the longissimus dorsi muscle, including its three sets of fasciculi.

Quick Recap

- Layer 4: Longissimus costarum (expiratory) + Longissimus dorsi (LONGEST/most powerful muscle in the body; extends/flexes back)
- Layer 5: Levatores costarum (inspiratory, 10-11 pairs) + Multifidus dorsi (extends back, continuous neck-to-tail chain) + Intertransversales lumborum (lateral flexion)

MYOLOGY — BACK, LOIN & TAIL

Muscles of the Tail

Overview

- Situated around the tail, enclosed in strong coccygeal fascia
- Tail movements from **5 pairs of coccygeal muscles** — 3 of these run LONGITUDINALLY around the coccygeal vertebrae, covering them completely
- Composed of fasciculi series terminating in tendons; gradually thin out (diminish) moving backward along the tail
- 6 named muscles total: Sacro-coccygeus dorsalis, Sacro-coccygeus lateralis, Sacro-coccygeus ventralis, Intertransversales caudae, Coccygeus, Recto-coccygeus
- The 3 "sacro-coccygeus" muscles are distinguished by position: **dorsalis, lateralis, ventralis**

Muscle	Also Called	Origin	Insertion	Action
Sacro-coccygeus dorsalis	Erector coccygeus	Last 3-4 sacral spines + some coccygeal vertebrae spines	Upper faces, coccygeal vertebral bodies (beyond the 3rd)	Elevate the tail, incline it laterally
Sacro-coccygeus lateralis	Curvator coccygeus	Spinous process, last lumbar + transverse process, sacral/coccygeal vertebrae	Lateral vertebral faces (except first 3)	Elevate the tail, flex it laterally
Sacro-coccygeus ventralis	Depressor coccygeus	Transverse process, 5th sacral vertebra (medial + lateral parts)	Medial part: coccygeal bodies. Lateral part: coccygeal bodies + transverse processes	DEPRESS the tail

Muscle	Origin	Insertion	Note
Intertransversales caudae	Sacrum's lateral border + preceding coccygeal vertebra's transverse process	Transverse process, 1st coccygeal + the vertebra behind	Lateral face of tail, between lateralis + ventralis muscles; fills inter-transverse spaces
Coccygeus	Pelvic sciatic ligament surface, near ischiatic spine	Transverse processes, 2nd + 3rd coccygeal vertebrae	Thin, triangular; against the pelvis's lateral wall (between sacro-sciatic ligament + rectum)
Recto-coccygeus	Detached (either side) from rectum's longitudinal muscular coat	4th + 5th coccygeal vertebrae	Large band of PLAIN (smooth) muscle

- Recto-coccygeus action: compress the anus, depress the tail over the perineum, flex the tail laterally
- Recto-coccygeus blood/nerve: Middle + lateral coccygeal arteries; Coccygeal nerves

IMPORTANT

Recto-coccygeus is unique among tail muscles — it's made of PLAIN (smooth/involuntary) muscle, not skeletal muscle, reflecting its close functional relationship to the rectum's own smooth muscle coat.

MNEMONIC

"Dorsalis Elevates, Lateralis curves+elevates, Ventralis depresses" — the 3 sacro-coccygeus muscles' actions follow their names' positions (dorsal=up, ventral=down).

Species Differences

Species	Key Feature
Horse	Resemble ox in general; coccygeus muscle is LESS developed
Dog	Resemble ox in general; coccygeal muscles participate in forming the PELVIC DIAPHRAGM; has an ADDITIONAL sacro-coccygeus arising from the sacrum's edge + 1st coccygeal transverse process + ilium's posterior border; sacro-coccygeus also arises from the lumbar vertebrae

IMPORTANT

In the dog, the coccygeal muscles contribute to forming the PELVIC DIAPHRAGM — directly relevant to canine perineal hernia, a common clinical condition linked to pelvic diaphragm muscle weakness.

PYQ

Which tail muscle is composed of plain (smooth) muscle rather than skeletal muscle?

A) Coccygeus **B) Recto-coccygeus ✓** C) Sacro-coccygeus dorsalis D) Intertransversales caudae

Exp: Recto-coccygeus is a band of plain muscle, detached from the rectum's own smooth muscular coat.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the three sacro-coccygeus muscles and their actions on the tail.

Quick Recap

- 6 tail muscles: Sacro-coccygeus dorsalis (elevate), lateralis (elevate+flex), ventralis (depress) + Intertransversales caudae + Coccygeus + Recto-coccygeus (plain muscle, compresses anus)
- Dog's coccygeal muscles help form the pelvic diaphragm (relevant to perineal hernia)

MYOLOGY — HEAD: FACIAL REGION (LIPS & CHEEKS)

Facial Region — Superficial Muscle & Lips/cheeks Muscles

Head Muscle Regions — Overview

- Head muscles surround: mouth, nostrils, eyelids, external ear, mandible, hyoid bones
- Sub-regions: **Facial** (superficial/cutaneous + lips/cheeks + nostrils) | **Palpebral** (eyelids) | **Auricular** (external ear) | **Temporo-maxillary** (mandibular/mastication) | **Hyoid** | **Orbital** (eyeball/ocular)

Superficial (Cutaneous) Muscle of the Face

- Well-developed, extensive; covers the whole frontal region = **frontalis muscle**
- The part extending over the lower jaw + masseter + buccinator, blending with orbicularis oris at the mouth angle = **retractor anguli oris**

Muscles of the Lips & Cheeks — Full List (12 Muscles)

- Orbicularis oris | Levator nasolabialis | Levator labii superioris proprius | Levator labii superioris proprius accessorius | Dilator naris lateralis (Caninus) | Depressor labii superioris | Zygomaticus | Incisivus superior | Incisivus inferior (levator labii inferioris) | Mentalis | Buccinator | Levator buccalis | Depressor labii inferioris

Muscle	Origin	Insertion	Action	Nerve
Orbicularis oris	— (sphincter, incomplete in muzzle)	Blends with all facial muscles converging at mouth angle	Close the lips, help suction + mastication	Facial nerve

Muscle	Origin	Insertion	Action	Nerve
Levator nasolabialis	Frontal + nasal bones (splits into superficial + deep layers below)	Upper lip + lateral nostril part	Elevate upper lip, dilate nostril	Facial nerve
Levator labii superioris proprius	Facial tuberosity	Several tendons, upper lip	Elevate upper lip / draw it to a side	Facial nerve
Levator labii superioris proprius accessorius	In front of facial tuberosity	Upper lip (several tendons)	ASSIST the preceding muscle	Facial nerve
Dilator naris lateralis (Caninus)	Facial tuberosity	Lateral wing of nostril	Dilate the nostril	Facial nerve
Depressor labii superioris	Facial tuberosity (below preceding)	Muzzle + upper lip (several tendons)	Compress the upper lip	Facial nerve
Zygomaticus	Zygomatic process, malar bone	Buccinator + orbicularis, at mouth angle	Retract + raise the mouth angle	Facial nerve
Incisivus superior	Antero-lateral angle, premaxilla	Deep surface, upper lip	Pull lip toward the premaxilla	Facial nerve
Incisivus inferior (Levator labii inferioris)	Mandible body, at 2nd incisor level	Lower lip	Elevate the chin	Facial nerve
Mentalis	Mandible body, either side of symphysis	Skin of the chin	Wrinkle skin, pull it forward	Mental nerve
Buccinator	Alveolar border, mandible + lower external maxilla surface	Mouth angle, blends with orbicularis	Mastication: flatten cheek, press food between teeth, retract mouth angle	Facial nerve
Levator buccalis	Facial part, lacrimal bone	Buccal mucosa, ventral to facial tuberosity	Raise posterior cheek	Superior buccal + sub-zygomatic nerve
Depressor labii inferioris	Alveolar mandible border, behind last cheek tooth	Lower lip	Depress the lower lip	Facial nerve

IMPORTANT

Almost ALL facial muscles here share the SAME nerve supply — the Facial nerve (VII). Only Mentalis (Mental nerve) and Levator buccalis (Superior buccal/sub-zygomatic nerve) are exceptions — a high-yield "spot the odd one out" fact.

EXAM TIP

Naming pattern to notice — "Levator" = raises something, "Depressor" = lowers something, "Dilator" = widens something. Once you know this prefix system, you can predict most of these muscles' actions without memorizing each individually.

MNEMONIC

For the upper-lip elevator trio — "Levator labii superioris proprius" + its "accessorius" + "Levator nasolabialis" — remember they all work TOGETHER to lift the upper lip, just from slightly different origins (facial tuberosity vs frontal/nasal bones).

PYQ

The muscle blending with the orbicularis oris at the angle of the mouth, formed by the lower-jaw extension of the cutaneous muscle, is called the?

A) Buccinator **B) Retractor anguli oris ✓** C) Zygomaticus D) Levator buccalis

Exp: The retractor anguli oris is the part of the cutaneous facial muscle extending over the lower jaw, masseter, and buccinator, blending with orbicularis oris at the mouth's angle.

PYQ

Which muscle is primarily responsible for mastication by flattening the cheek and pressing food between the teeth?

A) Zygomaticus **B) Buccinator ✓** C) Orbicularis oris D) Mentalis

Exp: Buccinator forms the main muscular mass of the cheek, and its chief action is in mastication.

PYQ

The mentalis muscle is supplied by which nerve, unlike most other facial muscles which use the facial nerve?

A) Mental nerve ✓ B) Trigeminal nerve C) Buccal nerve D) Infraorbital nerve

Exp: Mentalis is supplied by the mental nerve, an exception to the facial-nerve pattern seen in most lip/cheek muscles.

PYQ

The incisivus superior muscle pulls the upper lip toward which bone?

A) Maxilla B) Nasal bone **C) Premaxilla ✓** D) Mandible

Exp: Incisivus superior arises from the premaxilla's antero-lateral angle and pulls the lip toward it.

PYQ

Which muscle is described as "assisting" the levator labii superioris proprius, lying between it and the dilator naris lateralis?

A) Zygomaticus **B) Levator labii superioris proprius accessorius ✓** C) Depressor labii superioris D) Buccinator

Exp: The levator labii superioris proprius accessorius is regarded as accessory to the levator labii superioris proprius, positioned between it and the dilator naris lateralis.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the muscles of the lips and cheeks with their nerve supply, noting any exceptions to the facial nerve pattern.

Quick Recap

- 12 lip/cheek muscles — almost ALL supplied by Facial nerve, EXCEPT Mentalis (mental nerve) and Levator buccalis (superior buccal/sub-zygomatic nerve)
- Naming pattern: Levator=raises, Depressor=lowers, Dilator=widens — predicts most actions
- Buccinator = the main mastication muscle of the cheek; Retractor anguli oris = cutaneous muscle's mouth-angle extension

MYOLOGY — HEAD: FACIAL REGION (NOSTRILS)

Facial Region — Muscles of the Nostrils

Full List (5 Muscles — All Dilate the Nostril)

- Levator naso labialis (already covered under lips/cheeks — dual-listed since it also dilates the nostril) | Dilator naris lateralis (Caninus, already covered) | Dilator naris apicalis | Dilator naris superioris (dorsal part) | Dilator naris inferioris (ventral part)

Muscle	Origin	Insertion	Action	Blood/Nerve
Levator naso labialis	Frontal + nasal bones (2 layers: superficial + deep)	Upper lip + lateral nostril part	Elevate upper lip, dilate nostril	Facial, dorsal, labial, malar arteries; Facial nerve

Muscle	Origin	Insertion	Action	Blood/Nerve
Dilator naris lateralis (Caninus)	Facial tuberosity	Lateral wing of nostril	Dilate the nostril	Superior labial artery; Facial nerve
Dilator naris apicalis	Body of the premaxilla	Medial wing of nostril	Dilate the nostril	Naso-labial arteries; Facial nerve
Dilator naris superioris (dorsal part)	Alar cartilage of nostril	Medial wings of nostril	Dilate the nostril	Facial + infra-orbital arteries; Facial + infraorbital nerves
Dilator naris inferioris (ventral part)	Nasal process of premaxilla (arranged in 2 layers)	Lateral wing of nostril	Dilate the nostril	Facial + infra-orbital arteries; Facial + infraorbital nerves

IMPORTANT

EVERY single nostril muscle's action is simply "dilate the nostril" — the differentiation between these 5 muscles is purely about WHICH PART of the nostril they attach to (medial wing vs lateral wing vs apex) and their exact origin, not their functional action. A classic case where memorizing the ORIGIN + insertion matters far more than the action.

EXAM TIP

Two muscles ("Superioris" and "Inferioris") share the SAME dual nerve supply (Facial + Infraorbital nerves) — these are the only 2 nostril muscles with a combined nerve supply; the other 3 use Facial nerve alone.

MNEMONIC

"Apicalis at the tip, Superioris on top, Inferioris below, Lateralis to the side" — matches each dilator muscle name to its nostril position directly.

PYQ

How many distinct muscles act to dilate the nostril in the ox?

- A) 3 B) 4 **C) 5 ✓** D) 6

Exp: Five muscles dilate the nostril: levator naso labialis, dilator naris lateralis, apicalis, superioris, and inferioris.

PYQ

Which nostril dilator muscle arises specifically from the alar cartilage of the nostril itself?

- A) Dilator naris apicalis **B) Dilator naris superioris ✓** C) Dilator naris inferioris D) Dilator naris lateralis

Exp: Dilator naris superioris (dorsal part) uniquely arises from the alar cartilage of the nostril, unlike the others which arise from bone.

PYQ

The dilator naris apicalis, situated in the muzzle, arises from which bone?

- A) Nasal bone B) Frontal bone **C) Body of the premaxilla ✓** D) Maxilla

Exp: Dilator naris apicalis arises from the body of the premaxilla.

PYQ

Which two nostril muscles share a combined Facial + Infraorbital nerve supply, unlike the other three which use only the facial nerve?

- A) Apicalis and Lateralis **B) Superioris and Inferioris ✓** C) Naso labialis and Apicalis D) Lateralis and Superioris

Exp: Dilator naris superioris and inferioris both receive combined facial + infraorbital nerve supply, unlike the other three nostril muscles.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the five muscles of the nostrils, noting that all share the same basic action.

Quick Recap

- 5 nostril muscles, ALL dilate the nostril — differ only in origin/insertion location (apex/dorsal/ventral/lateral)
- Nerve supply: mostly Facial nerve alone; ONLY Superioris + Inferioris get combined Facial + Infraorbital supply
- Levator nasolabialis + Dilator naris lateralis are dual-purpose (also lift the upper lip / seen in the lips-cheeks list too)

MYOLOGY — HEAD: PALPEBRAL REGION

Palpebral Region — Muscles of the Eyelids

Full List

- Orbicularis oculi | Corrugator supercilli | Malaris (splits into: Levator buccalis + Depressor palpebrae inferioris) | Levator palpebrae superioris

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Orbicularis oculi	Well-developed, flat, elliptical SPHINCTER around the eyelids; fibres chiefly attach to skin + palpebral ligament at medial canthus	—	—	Close the eyelids	Alar + frontal arteries; Facial nerve
Corrugator supercilli	In the OX, this muscle's "place is taken up" by the lower part of the frontalis muscle (around the upper eyelid) — i.e. not a fully separate distinct muscle in ox	—	—	Raise the upper eyelid, wrinkle the skin	Frontal arteries; Facial nerve
Malaris	Broad muscle spreading below the lower eyelid, over buccinator + masseter; divides into an anterior part (levator buccalis) + posterior part (depressor palpebrae inferioris)	—	—	(see 2 divisions below)	—

Malaris — Its Two Divisions

Division	Origin	Insertion	Action	Blood/Nerve
Levator buccalis	Facial part, lacrimal bone	Buccal mucosa, ventral to facial tuberosity	Raise the posterior cheek	Malar branch, external maxillary artery; Superior buccal + sub-zygomatic nerve
Depressor palpebrae inferioris	Facial part, lacrimal bone	Fascia, dorsal buccinator border	Attaches to lower eyelid; DEPRESSES the lower eyelid	Branch from external maxillary artery; Superior buccal + sub-zygomatic nerve

Levator Palpebrae Superioris

Feature	Detail
Location	Within the orbit, above the dorsal rectus muscle
Origin	Pterygoid crest, opposite the optic foramen
Insertion	Upper eyelid
Action	Raise the upper eyelid
Blood supply	Ophthalmic artery
Nerve supply	Oculomotor nerve

IMPORTANT

In the OX, corrugator supercilli is NOT a fully independent muscle — its role is taken over by the lower part of the frontalis muscle around the upper eyelid. This is a genuinely tricky species-specific nuance worth flagging clearly, since it could easily be mistested as "corrugator supercilli is absent in ox" when the more accurate statement is that a different muscle performs its role.

EXAM TIP

Notice the paired opposite-action structure: Levator buccalis RAISES the cheek/lower lid area, while Depressor palpebrae inferioris (Malaris's other half) LOWERS the eyelid — same parent muscle (Malaris), opposite directional actions from its two divisions.

MNEMONIC

"Oculi closes, Supercilli/Frontalis raises the brow, Malaris splits up-cheek/down-lid, Superioris lifts the lid" — a quick chain covering all 4 eyelid muscle groups' core actions.

PYQ

Which muscle acts as a sphincter to close the eyelids?

- A) Levator palpebrae superioris **B) Orbicularis oculi ✓** C) Corrugator supercilli D) Malaris

Exp: Orbicularis oculi is the flat, elliptical sphincter muscle surrounding and closing the eyelids.

PYQ

In the ox, the role of corrugator supercilli is effectively taken over by which muscle?

- A) Orbicularis oculi B) Levator palpebrae superioris **C) The lower part of the frontalis muscle ✓** D) Malaris

Exp: In the ox, corrugator supercilli's function around the upper eyelid is performed by the lower part of the frontalis muscle instead of a fully separate muscle.

PYQ

The levator palpebrae superioris is supplied by which cranial nerve?

A) Facial nerve B) Trigeminal nerve **C) Oculomotor nerve ✓** D) Abducent nerve

Exp: Levator palpebrae superioris is supplied by the oculomotor nerve (III), unlike most other eyelid muscles which use the facial nerve.

PYQ

The malaris muscle divides into which two named parts?

A) Orbicularis and corrugator **B) Levator buccalis and depressor palpebrae inferioris ✓** C) Superioris and inferioris D) Dorsalis and ventralis

Exp: Malaris divides into an anterior part (levator buccalis) and a posterior part (depressor palpebrae inferioris).

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the malaris muscle and its two divisions with their actions.

Quick Recap

- Orbicularis oculi = sphincter, closes eyelids (Facial nerve)
- Corrugator supercilli = raises upper eyelid; in OX, replaced functionally by lower frontalis (not a separate distinct muscle here)
- Malaris = splits into Levator buccalis (raises cheek) + Depressor palpebrae inferioris (lowers eyelid)
- Levator palpebrae superioris = raises upper eyelid; UNIQUE nerve supply = Oculomotor (III), not facial nerve like the rest

MYOLOGY — HEAD: AURICULAR REGION

Auricular Region — Extrinsic & Intrinsic Muscles

Overview — Two Groups

Group	Definition	Muscles
Extrinsic	Arise from EXTRANEIOUS parts (cranial bones, ligamentum nuchae, parotid gland) → insert into ear cartilage	Parotido-auricularis, Cervico-auricularis, Parieto-auricularis externus, Parieto-auricularis internus, Zygomatico-auricularis, Temporo-auricularis (Tragicus in horse)
Intrinsic	CONFINED entirely to the ear cartilage itself	Scuto-auricularis externus, Scuto-auricularis internus

Extrinsic Muscles — Detail Table

Muscle	Description	Origin	Insertion	Action
Parotido-auricularis	Thin, wide; on the parotid salivary gland; wide below, narrow above	Fascia, lower parotid gland part	Conchal cartilage, below the border-junction angle	Incline the ear downward + backward
Cervico-auricularis	COMPOUND muscle — 3 parts (externus, medius, internus), named by position at origin; arranged layer by layer	Ligamentum nuchae + cervical fascia	Medial face, concha's base	Rotate the ear, draw it backward

Muscle	Description	Origin	Insertion	Action
Parieto-auricularis externus	Wide but THIN; covers the temporalis	Frontal crest + parietal + occipital bones	Scutiform cartilage + anterior concha aspect	Adduct the concha, rotate it forward (to "prick" the ear)
Parieto-auricularis internus	Placed PARTIALLY under the preceding muscle	Parietal bone, behind the frontal eminence	Medial face of the concha	Adduct the ear
Zygomatico-auricularis	Made of TWO bundles	Zygomatic process, squamous temporal	Upper bundle: scutiform cartilage. Lower bundle: front + base of concha	Prick the ear
Temporo-auricularis (Tragicus of horse)	Very thin fasciculus	External auditory process + annular cartilage	Base of concha	Telescope the conchal + annular cartilage, SHORTEN the external acoustic meatus

Intrinsic Muscles — Detail Table

Muscle	Origin	Insertion	Action
Scuto-auricularis externus	Outer face, scutiform cartilage	Medial part, concha's base	Prick the ear
Scuto-auricularis internus	Internal face, scutiform cartilage	Anterior aspect of the concha	Prick the ear

- Blood supply (both intrinsic muscles): External carotid + superficial temporal arteries + a small occipital artery branch
- Nerve supply (both intrinsic muscles): Facial nerve + 1st/2nd spinal nerves

IMPORTANT

NOTICE how many of these ear muscles share the exact same action — "to prick the ear" — appearing for Zygomatico-auricularis, Scuto-auricularis externus, AND Scuto-auricularis internus. This redundancy across 3+ muscles for the same basic movement is itself worth remembering as a pattern (the ear-pricking reflex uses multiple cooperating muscles, not just one).

EXAM TIP

"Scutiform cartilage" is the shared anatomical landmark connecting MULTIPLE ear muscles (Parieto-auricularis externus, Zygomatico-auricularis, BOTH Scuto-auricularis muscles) — recognizing this one structure as a common attachment hub links together nearly half the ear muscle list.

MNEMONIC

"Parotido pulls DOWN-BACK, Cervico ROTATES-BACK, Parieto ADDUCTS, Zygomatico + Scuto PRICK" — a rough action-grouping across the 6 extrinsic + 2 intrinsic muscles.

PYQ

The compound muscle with three named parts (externus, medius, internus), arising from the ligamentum nuchae and cervical fascia, is the?

A) Parotido-auricularis **B) Cervico-auricularis ✓** C) Parieto-auricularis D) Zygomatico-auricularis

Exp: Cervico-auricularis is a compound muscle with three parts (externus, medius, internus), arranged in layers, arising from the ligamentum nuchae and cervical fascia.

PYQ

Which ear muscle's specific action is to shorten the external acoustic meatus by telescoping the conchal and annular cartilages?

A) Parotido-auricularis **B) Temporo-auricularis (Tragicus) ✓** C) Scuto-auricularis internus D) Zygomatico-auricularis

Exp: Temporo-auricularis (called Tragicus in the horse) telescopes the conchal and annular cartilage, shortening the external acoustic meatus.

PYQ

The scutiform cartilage serves as an insertion or origin point for how many of the named ear muscles listed?

A) 1 B) 2 **C) 3 or more ✓** D) None

Exp: The scutiform cartilage is shared by parieto-auricularis externus, zygomatico-auricularis (upper bundle), and both scuto-auricularis muscles — at least 4 muscles reference it.

PYQ

Intrinsic ear muscles are distinguished from extrinsic ones because they?

A) Are larger B) Arise from cranial bones **C) Are confined entirely to the ear cartilage ✓** D) Only exist in the horse

Exp: Intrinsic muscles (the two scuto-auricularis muscles) are confined entirely to the ear's own cartilage, unlike extrinsic muscles which arise from bones/fascia/glands outside the ear.

PYQ

Which extrinsic ear muscle originates from the parotid salivary gland's fascia?

A) Cervico-auricularis **B) Parotido-auricularis ✓** C) Parieto-auricularis internus D) Temporo-auricularis

Exp: Parotido-auricularis arises from the fascia on the lower part of the parotid gland, as its name suggests.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Classify the muscles of the external ear into extrinsic and intrinsic groups, with their basic actions.

Quick Recap

- Extrinsic (from bones/fascia/gland): Parotido-auricularis, Cervico-auricularis (3-part compound), Parieto-auricularis (ext+int), Zygomatico-auricularis, Temporo-auricularis
- Intrinsic (confined to ear cartilage only): Scuto-auricularis externus + internus
- Scutiform cartilage = shared hub for 4+ muscles; "prick the ear" = shared action for 3+ muscles
- Temporo-auricularis (Tragicus in horse) = uniquely shortens the external acoustic meatus

MYOLOGY — HEAD: TEMPORO-MANDIBULAR REGION

Muscles of Mastication

Overview

- Also called: mandibular muscles / muscles of mastication
- List: Masseter | Temporalis | Pterygoideus medialis | Pterygoideus lateralis | Digastricus

Muscle	Origin	Insertion	Action	Blood/Nerve
Masseter	Zygomatic arch, facial crest, facial tuberosity	Lateral face + posterior border, mandible's vertical ramus	CLOSE the jaws, move mandible forward-upward; SINGLY = lateral jaw motion	Transverse facial, masseteric, external maxillary branch, deep temporal arteries; Mandibular nerve
Temporalis	Rough part of the temporal fossa	Coronoid process of the mandible	RAISE the lower jaw	Deep temporal artery; Mandibular nerve

Muscle	Origin	Insertion	Action	Blood/Nerve
Pterygoideus medialis	Lateral face, palatine's perpendicular part + sphenoid's pterygoid crest	Medial surface, mandible's vertical ramus (behind the mandibular foramen)	RAISE the lower jaw; SINGLY assists lateral jaw movement	1-2 pterygoid arteries (from internal maxillary); Mandibular nerve
Pterygoideus lateralis	Pterygoid process, sphenoid bone	Medial face, mandible's neck + condyle + the articular disc	Draw the jaw FORWARD; SINGLY moves it to the opposite side	Branch from mandibular artery; Mandibular nerve
Digastricus	Paramastoid process	Medial face, horizontal ramus (near ventral border)	DEPRESS the lower jaw	Branch from external maxillary artery; Facial + Mandibular nerves

- Digastricus: LONG muscle, 2 muscular bellies + an intermediate tendon; the 2 anterior bellies connect via a band below the mylohyoideus = **transversus mandibulae**

IMPORTANT

All 4 "closing/raising" jaw muscles (Masseter, Temporalis, Pterygoideus medialis, Pterygoideus lateralis) are supplied by the SAME nerve — the Mandibular nerve (a branch of Trigeminal/V). Only Digastricus (the JAW-OPENER) breaks this pattern, using BOTH Facial AND Mandibular nerves.

EXAM TIP

Simple functional split — Masseter + Temporalis + Pterygoideus medialis = CLOSE/RAISE the jaw; Digastricus = OPENS/DEPRESSES the jaw; Pterygoideus lateralis = uniquely draws the jaw FORWARD (protrusion) rather than up/down.

MNEMONIC

"MTP close, D opens, L(ateral) protrudes" — Masseter-Temporalis-Pterygoideus(medial) CLOSE the jaw; Digastricus opens; Pterygoideus Lateral pulls it forward.

PYQ

Which muscle of mastication is responsible for depressing (opening) the lower jaw?

- A) Masseter B) Temporalis **C) Digastricus ✓** D) Pterygoideus medialis

Exp: Digastricus is the jaw-opener/depressor; the other three (masseter, temporalis, pterygoideus medialis) all close/raise the jaw.

PYQ

The transversus mandibulae is the connecting band between which structures?

- A) The two temporalis muscles **B) The two anterior bellies of digastricus ✓** C) Masseter and temporalis D) Pterygoideus medialis and lateralis

Exp: Transversus mandibulae connects the two anterior bellies of digastricus, situated below the mylohyoideus.

PYQ

Which muscle of mastication uniquely draws the jaw forward rather than simply opening or closing it?

- A) Masseter **B) Pterygoideus lateralis ✓** C) Temporalis D) Digastricus

Exp: Pterygoideus lateralis's specific action is to draw the jaw forward; acting singly it moves the jaw to the opposite side.

PYQ

Digastricus is unique among the muscles of mastication because it is supplied by?

- A) Only the mandibular nerve B) Only the facial nerve **C) Both facial and mandibular nerves ✓** D) The trigeminal nerve alone

Exp: Digastricus receives supply from BOTH facial and mandibular nerves, unlike the other 4 mastication muscles which use only the mandibular nerve.

PYQ

The temporalis muscle originates from which structure?

A) Zygomatic arch **B) Rough part of the temporal fossa ✓** C) Pterygoid process D) Paramastoid process

Exp: Temporalis arises from the rough part of the temporal fossa and inserts on the coronoid process of the mandible.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the five muscles of mastication with their actions, noting which one uniquely opens the jaw.

Quick Recap

- 5 mastication muscles: Masseter + Temporalis + Pterygoideus medialis (CLOSE jaw) | Pterygoideus lateralis (draws jaw FORWARD) | Digastricus (OPENS/depresses jaw)
- Nerve pattern: all 4 closers use Mandibular nerve alone; Digastricus uniquely uses BOTH Facial + Mandibular nerves
- Transversus mandibulae = band connecting digastricus's 2 anterior bellies

MYOLOGY — HEAD: HYOID REGION

Muscles of the Hyoid Region

Overview

- 8 muscles: Mylo-hyoideus | Stylo-hyoideus | Genio-hyoideus | Occipito-hyoideus | Kerato-hyoideus | Hyoideus transversus | Sterno-thyro-hyoideus | Omo-hyoideus

Muscle	Origin	Insertion	Action	Blood/Nerve
Mylo-hyoideus	Medial mandible face, below the alveolar bone	Medial fibrous raphe + lingual process of hyoid	Raise the tongue + hyoid bone	Sublingual artery; Mandibular nerve
Stylo-hyoideus	Muscular angle, great cornu of hyoid	Lateral extremity, hyoid body	Draw tongue + hyoid bone UP and BACK	External maxillary artery branch; Facial nerve (styloid branch)
Genio-hyoideus	Fossa, medial mandible ramus face, close to symphysis	Hyoid body	Draw tongue + hyoid FORWARD	Lingual + sublingual artery; Hypoglossal nerve
Occipito-hyoideus	Paramastoid process	Posterior border, stylohyoid's muscular angle	Draw the great cornu BACKWARD	Posterior auricular artery; Facial nerve
Kerato-hyoideus	Posterior border, small cornu	Dorsal border, thyroid cornu	RAISE the larynx	Lingual artery; Glosso-pharyngeal nerve
Hyoideus transversus	(unpaired; between the 2 small cornua, enclosed by mucous membrane)	—	Raise the root of the tongue	Lingual artery; Glosso-pharyngeal nerve
Sterno-thyro-hyoideus	Manubrium sterni (2 side muscles close together, cover trachea's lower 1/3; each divides mid-neck into thyroid + hyoid bands)	Thyroid band: thyroid cartilage of larynx. Hyoid band: hyoid body + lingual process	Depress the hyoid, larynx, and tongue during DEGLUTITION (swallowing)	Carotid artery; 1st + 2nd cervical spinal nerves

Muscle	Origin	Insertion	Action	Blood/Nerve
Omo-hyoideus	Transverse processes, C3 + C4 (passes under brachiocephalicus/sterno-mastoideus, obliquely forward-inward)	Hyoid bone body	RETRACT the hyoid + tongue root	Carotid artery; 2nd cervical spinal nerve

- Mylo-hyoideus: the two muscles TOGETHER form a sling for the tongue
- Sterno-thyro-hyoideus: in the neck's UPPER 1/3, the trachea's ventral face is NOT covered by any muscle (since the bands diverge before reaching there) — thyroid band = LARGER + more lateral of the two bands

IMPORTANT

Kerato-hyoideus and Hyoideus transversus are the ONLY two hyoid muscles supplied by the Glosso-pharyngeal nerve — every other hyoid muscle uses a different nerve (Mandibular, Facial, Hypoglossal, or cervical spinal nerves). A useful "spot the shared exception pair" fact.

EXAM TIP

Genio-hyoideus pulls the tongue/hyoid FORWARD while Occipito-hyoideus pulls the great cornu BACKWARD — directly opposite actions, useful as a paired contrast.

MNEMONIC

"Mylo raises (sling), Stylo up-back, Genio forward, Occipito back, Kerato raises larynx, Transversus raises tongue root, Sterno-thyro-hyoid depresses (swallowing), Omo retracts" — one action word per muscle, in order.

PYQ

Which hyoid muscle acts specifically during deglutition (swallowing) to depress the hyoid, larynx, and tongue together?

- A) Genio-hyoideus **B) Sterno-thyro-hyoideus ✓** C) Omo-hyoideus D) Mylo-hyoideus

Exp: Sterno-thyro-hyoideus depresses the hyoid, larynx, and tongue during deglutition.

PYQ

The two muscles of the hyoid region supplied by the glosso-pharyngeal nerve are?

- A) Mylo-hyoideus and Stylo-hyoideus **B) Kerato-hyoideus and Hyoideus transversus ✓** C) Genio-hyoideus and Occipito-hyoideus D) Sterno-thyro-hyoideus and Omo-hyoideus

Exp: Kerato-hyoideus and hyoideus transversus are both supplied by the glosso-pharyngeal nerve, unlike the other hyoid muscles.

PYQ

Mylo-hyoideus, forming a sling for the tongue, is supplied by which nerve?

- A) Hypoglossal nerve B) Facial nerve **C) Mandibular nerve ✓** D) Glosso-pharyngeal nerve

Exp: Mylo-hyoideus is supplied by the mandibular nerve.

PYQ

Which hyoid muscle is unpaired, lying between the two small cornua enclosed by mucous membrane?

- A) Kerato-hyoideus **B) Hyoideus transversus ✓** C) Genio-hyoideus D) Stylo-hyoideus

Exp: Hyoideus transversus is the single, unpaired muscle situated between the two small cornua.

PYQ

The genio-hyoideus is supplied by which nerve, distinguishing it from most other hyoid muscles?

- A) Facial nerve **B) Hypoglossal nerve ✓** C) Glosso-pharyngeal nerve D) Mandibular nerve

Exp: Genio-hyoideus is supplied by the hypoglossal nerve, unique among this muscle group.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the eight muscles of the hyoid region with their actions during swallowing.

Quick Recap

- 8 hyoid muscles, each with a DIFFERENT primary nerve (Mandibular, Facial, Hypoglossal, Glosso-pharyngeal x2, Cervical spinal x2) — nerve supply is highly muscle-specific here, unlike the mastication group
- Mylo-hyoideus = tongue sling (raises tongue+hyoid); Sterno-thyro-hyoideus = key swallowing muscle (depresses hyoid/larynx/tongue)
- Genio-hyoideus (forward) vs Occipito-hyoideus (backward) = opposite-direction pair

MYOLOGY — HEAD: EYEBALL (OCULAR MUSCLES)

Muscles of the Eyeball

Overview

- Enclosed in superficial + deep fasciae
- **7 muscles total:** 4 Straight (Recti) + 1 Retractor + 2 Oblique
- List: Rectus superioris (dorsalis) | Rectus inferioris (ventralis) | Rectus medialis | Rectus lateralis | Retractor oculi | Oblique superior | Oblique inferior

Straight (Rectus) Muscles — 4 Muscles

- Named BOTH by position (superioris/dorsalis, inferioris/ventralis, medialis, lateralis) AND by action (levator, depressor, adductor, abductor oculi)
- Flat muscular bands; arise around the optic foramen, diverge, pass forward along corresponding eyeball aspects, terminate in fine aponeurotic tendons inserted to the sclera's fore part (in front of the equator)

Muscle	Action
Rectus superioris (dorsalis)	Elevates the eyeball / turns it upward = "Levator oculi"
Rectus inferioris (ventralis)	Depresses the eyeball / turns it downward = "Depressor oculi"
Rectus lateralis (external rectus)	Abducts the eyeball / turns it outward = "Abductor oculi"
Rectus medialis (internal rectus)	Adducts the eyeball / turns it inward = "Adductor oculi"

- ALL 4 recti acting TOGETHER = retract the eyeball

EXAM TIP

Notice the dual naming system — position-based names (superioris/inferioris/medialis/lateralis) map exactly to action-based names (levator/depressor/adductor/abductor). Learning one set lets you derive the other automatically.

Posterior Muscle — Retractor Oculi

Feature	Detail
Description	Enclosed WITHIN the recti muscles; envelops the optic nerve
Structure	4 bundles, arise around the optic foramen; bundles ALTERNATE with the recti (interposed between them)
Insertion	Posterior sclera part, BEHIND the recti
Action	Retract the eyeball into the orbit

IMPORTANT

Both the 4 recti (acting together) AND the retractor oculi (a separate 5th structure) can retract the eyeball — this is not a contradiction; the recti's combined pull plus the retractor's dedicated bundles work together for full retraction, but retractor oculi is the DEDICATED retraction muscle specifically.

Oblique Muscles — 2 Muscles

Muscle	Description	Origin	Course	Insertion	Action
Oblique superior	LONG; differs from others — has an INTERRUPTED course; narrow muscular band + thin aponeurotic termination	Back of the orbit, near the ethmoidal foramen	Forward along the orbit's internal wall (medial to internal rectus); at the supraorbital process level, gains a fibro-cartilaginous PULLEY, reflects around it outward-forward	Between the dorsal + lateral recti, ~half an inch behind the cornea's margin	Elevates AND rotates the eyeball
Oblique inferior	SHORTER + THICKER than superior; entirely muscular	Depression behind the lacrimal fossa	Curves around the rectus ventralis	Sclera, near/partly beneath the lateral rectus	Depresses AND draws the eyeball inward

- Blood supply (all 7 ocular muscles): Ophthalmic artery

Nerve Supply — The Key Exception Pattern

Nerve	Supplies
Oculomotor nerve	ALL ocular muscles EXCEPT the 3 listed below
Trochlear nerve	Superior oblique ONLY
Abducent nerve	Lateral rectus + Retractor oculi ONLY

IMPORTANT

This is one of THE most classic neuroanatomy bit-question patterns in veterinary/medical anatomy — Oculomotor supplies MOST eye muscles; Trochlear supplies ONLY superior oblique; Abducent supplies ONLY lateral rectus + retractor. Memorize the 2 EXCEPTIONS (trochlear, abducent) and everything else defaults to oculomotor.

MNEMONIC

"LR6 SO4 (the rest 3)" — Lateral Rectus = cranial nerve 6 (Abducent); Superior Oblique = cranial nerve 4 (Trochlear); everything else = cranial nerve 3 (Oculomotor). A widely-used classic mnemonic pattern applicable here too, extended to include retractor oculi under Abducent.

PYQ

How many muscles move the eyeball in total?

A) 5 B) 6 C) 7 ✓ D) 8

Exp: There are 7 ocular muscles: 4 recti, 1 retractor, and 2 obliques.

PYQ

Which cranial nerve supplies ONLY the superior oblique muscle?

A) Oculomotor **B) Trochlear ✓** C) Abducent D) Trigeminal

Exp: The trochlear nerve exclusively supplies the superior oblique muscle among the ocular muscles.

PYQ

The abducent nerve supplies which two ocular structures?

A) Superior oblique and inferior oblique **B) Lateral rectus and retractor oculi ✓** C) Medial rectus and superior rectus D) All four recti

Exp: The abducent nerve supplies the lateral rectus and the retractor oculi — the only two ocular muscles NOT supplied by oculomotor or trochlear.

PYQ

Which ocular muscle is unique in having an interrupted course, passing through a fibro-cartilaginous pulley near the supraorbital process?

A) Oblique inferior **B) Oblique superior ✓** C) Rectus lateralis D) Retractor oculi

Exp: Oblique superior has an interrupted course, reflecting around a fibro-cartilaginous pulley at the supraorbital process level before reaching its insertion.

PYQ

The retractor oculi muscle is made up of how many bundles, arranged alternating with the recti?

A) 2 B) 3 **C) 4 ✓** D) 6

Exp: Retractor oculi consists of 4 bundles arising around the optic foramen, alternating in position with the 4 recti muscles.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the nerve supply pattern of the ocular muscles, highlighting the two exceptions to oculomotor supply.

Quick Recap

- 7 ocular muscles: 4 Recti (superior/inferior/medial/lateral) + 1 Retractor oculi + 2 Oblique (superior/inferior)
- Nerve pattern: Oculomotor = most muscles | Trochlear = ONLY superior oblique | Abducent = ONLY lateral rectus + retractor oculi
- Oblique superior = interrupted course via a pulley; Oblique inferior = short, thick, entirely muscular
- All 4 recti together = retract the eyeball; Retractor oculi = the DEDICATED retraction muscle

MYOLOGY — HEAD: HORSE/DOG VARIATIONS

Muscles of the Head — Horse Vs Dog Variations

Horse — Key Differences from Ox

Region	Horse Variation
Facial (general)	Cutaneous muscle of the head LESS developed
Lips/mouth	Orbicularis oris = COMPLETE sphincter; Levator labii superioris proprius inserts by a common tendon with its fellow to the upper lip; Depressor labii superioris ABSENT
Nostrils	Transversus nasi = unpaired, lies BETWEEN the nostrils (a horse-specific muscle not in the ox list)
Eyelids	Corrugator supercilli = a fully SEPARATE muscle (unlike ox, where frontalis takes its role); Malaris thin; Levator buccalis ABSENT
Mandibular region	Extra muscle: Occipito-mandibularis — paramastoid process to the vertical ramus's posterior border

Region	Horse Variation
Digastricus	Has a DISTINCT intermediate tendon, which perforates the tendon of stylo-hyoideus
Thyro-arytenoideus (laryngeal)	Represented by ventricularis + vocalis (2 separate parts)
Other muscles	Resemble ox

Dog — Key Differences from Ox

Region	Dog Variation
Facial (general)	Facial cutaneous muscle WELL developed
Nostrils/lips	Levator naso labialis NOT divided; Levator labii superioris terminates by numerous small tendons (some unite with the opposite muscle, others insert into the nostrils); a triangular muscle below levator labii superioris represents dilator naris lateralis; OTHER dilator nostril muscles ABSENT; Depressor labii inferioris ABSENT
Eyelids	Corrugator supercilli extends from frontal bone fascia, terminates at the upper eyelid's medial angle
Mastication	Masseter shows REMARKABLE development — arises from zygomatic arch, prolonged beyond the mandibular ramus's posterior border, thick, convex external surface; Temporalis STRONGER (more tendinous material, larger temporal fossa); Lateral pterygoid NOT distinct from medial pterygoid
Mandibular region	Occipito-mandibularis = strong, round muscle (present here too, like horse)
Hyoid region	Stylo-hyoideus = very slender, pale muscular band; Kerato-hyoideus = LARGE; Occipito-hyoideus + Hyoideus transversus = BOTH ABSENT
Auricular (ear)	Parotido-auricularis LONGER + narrower; Parieto-auricularis externus + parieto-auricularis internus divide at termination into 2 parts (insert to both conchal + scutiform cartilage); Scuto-auricularis externus NOT distinct from parieto-auricularis externus
Other muscles	Resemble ox

IMPORTANT

Both horse AND dog have an "Occipito-mandibularis" muscle that does NOT exist in the ox's muscle list at all — a genuinely new muscle appearing in both other species but absent from the ox baseline.

EXAM TIP

Masseter's "remarkable development" in the dog (extending beyond the mandible's posterior border, unusually thick) is a distinctive, specific fact — useful for identifying dog skull/jaw specimens by muscle attachment markings alone.

PYQ

Which muscle, absent in the ox's muscle list entirely, appears in BOTH horse and dog?

A) Digastricus **B) Occipito-mandibularis ✓** C) Masseter D) Temporalis

Exp: Occipito-mandibularis is a new muscle present in both horse and dog but not listed for the ox.

PYQ

In the dog, which two hyoid muscles are both completely absent?

- A) Mylo-hyoideus and Stylo-hyoideus **B) Occipito-hyoideus and Hyoideus transversus ✓** C) Genio-hyoideus and Kerato-hyoideus D) Sterno-thyro-hyoideus and Omo-hyoideus

Exp: Occipito-hyoideus and hyoideus transversus are both absent in the dog.

PYQ

The transversus nasi muscle, unpaired and lying between the nostrils, is specific to which species?

- A) Ox **B) Horse ✓** C) Dog D) All three

Exp: Transversus nasi is a horse-specific muscle, unpaired, lying between the nostrils.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Compare the mandibular/mastication muscles of horse and dog, noting the occipito-mandibularis muscle present in both.

Quick Recap

- BOTH horse and dog have Occipito-mandibularis (absent in ox)
- Horse: Corrugator supercilli = fully separate muscle (unlike ox); has Transversus nasi (unique); Digastricus has a distinct perforating intermediate tendon
- Dog: Masseter = remarkably developed; Occipito-hyoideus + Hyoideus transversus = BOTH absent; ear muscles show extra terminal divisions

MYOLOGY — NECK: VENTRAL CERVICAL GROUP

Neck Muscles — Ventral Cervical Group

Overview

- Cutaneous muscle is ABSENT in the neck region
- Cervical muscles = 2 groups: **Ventral cervical** + **Lateral cervical**
- Ventral group lies below/lateral to cervical vertebrae, mostly around the trachea
- List: Sterno-cephalicus | Sterno-thyro-hyoideus (already covered under Larynx/Hyoid muscles) | Omo-hyoideus (already covered under Hyoid muscles) | Scalenus | Rectus capitis ventralis major (Longus capitis) | Rectus capitis ventralis minor (Rectus capitis ventralis) | Rectus capitis lateralis | Intertransversales colli | Longus colli

Sterno-Cephalicus

- 2 divisions: **Sterno-mandibularis** (superficial) + **Sterno-mastoideus** (deep)
- Both arise TOGETHER from the manubrium sterni + 1st costal cartilage; close together at origin/lower neck, diverge from mid-neck
- Sterno-mandibularis = superficial; forms the LOWER boundary of the jugular furrow
- Sterno-mastoideus = deep; covers oesophagus/trachea/carotid structures; crosses UNDER the superficial division at the upper 1/3 of neck

Feature	Detail
Insertion	Superficial division: anterior masseter border + jaw angle. Deep division: common with rectus capitis ventralis major, to mastoid process + basilar tubercles
Action	TOGETHER: flex head + neck. SINGLY: incline head + neck laterally
Blood supply	Common carotid artery

Feature	Detail
Nerve supply	Spinal accessory nerve

Scalenus

- 2 divisions (dorsal + ventral), between which the brachial plexus roots emerge
- Dorsal part (scalenus dorsalis): C4-C7 transverse processes; covered by serratus thoracis
- Ventral part (scalenus ventralis): C4-C6 transverse processes; phrenic nerve roots traverse it; brachial vessels cross its ventral edge

Feature	Detail
Insertion	Dorsal part: into the 4th rib. Ventral part: 1st rib's anterior border
Action	Flex the neck / incline it laterally. If neck fixed, acts as an INSPIRATORY muscle
Blood supply	Vertebral + intercostal arteries
Nerve supply	Cervical spinal nerves

Rectus Capitis Group (3 Muscles)

Muscle	Origin	Insertion	Action	Blood/Nerve
Rectus capitis ventralis major (Longus capitis)	Transverse processes, C2-C6	Basilar tubercle	Flex the head / incline it laterally	Vertebral + occipital arteries; Cervical spinal nerves
Rectus capitis ventralis minor (Rectus capitis ventralis)	Ventral surface, atlas wing's exterior	Basilar tubercle	Flex the occipito-atlantal joint	Occipital artery; 1st cervical spinal nerve
Rectus capitis lateralis	Atlas, lateral to obliquus capitis anterior's origin	Paramastoid process	Flex the occipito-atlantal joint	Occipital artery; 1st cervical spinal nerve

- Rectus capitis ventralis minor: SMALLER than the major; placed partly above/below/external to it
- Rectus capitis lateralis: SMALLER than the preceding; under cover of obliquus capitis anterior

Intertransversales Colli & Longus Colli

Muscle	Description	Attachment/Origin	Insertion	Action	Blood/Nerve
Intertransversales colli (Intertransversarii cervicis)	6 in number; fills space between articular + transverse processes; each has dorsal + ventral parts; origin/insertion NOT distinctly recognized	Dorsal bundles: upper transverse process to preceding vertebra's anterior articular process. Ventral bundles: between ventral transverse process parts; superficial fibres unite as "intertransversarius longus" (anterior attachment = atlas's lateral border)	—	Flex the neck laterally	Vertebral artery; Cervical spinal nerves (except 1st + last)
Longus colli	Long muscle, T6 to atlas; ventral aspect of bodies, closely applied to its fellow; thoracic + cervical parts	Bodies, first 6 thoracic vertebrae to medial surface, cervical vertebrae transverse processes	Thoracic part: bodies + transverse processes, last 2 cervical vertebrae. Cervical part: cervical vertebrae bodies + atlas's ventral tubercle	Flex the neck	Subcostal + vertebral arteries; Cervical + dorsal spinal nerves

IMPORTANT

Sterno-cephalicus is unique among this group in using the Spinal Accessory nerve — every other ventral cervical muscle listed here uses either cervical spinal nerves, vertebral/occipital-artery-paired nerves, or a mix, but NONE else use spinal accessory.

EXAM TIP

All 3 "rectus capitis" muscles (ventralis major, ventralis minor, lateralis) share the SAME basic action family — flexing the head/occipito-atlantal joint — differing mainly in size and precise origin, similar to the quadriceps femoris pattern (multiple muscles, same joint, same action, different sizes/origins).

PYQ

Which ventral cervical muscle is supplied by the spinal accessory nerve, unlike its neighbors?

- A) Scalenus B) **Sterno-cephalicus ✓** C) Longus colli D) Rectus capitis lateralis

Exp: Sterno-cephalicus is supplied by the spinal accessory nerve, distinct from the cervical-spinal-nerve pattern of most other ventral cervical muscles.

PYQ

The scalenus muscle, when the neck is fixed, can additionally act as a(n)?

- A) Expiratory muscle B) **Inspiratory muscle ✓** C) Postural stabilizer only D) None of these

Exp: When the neck is fixed, scalenus acts as an inspiratory muscle in addition to its usual neck-flexing role.

PYQ

Longus colli extends from which two points?

- A) Atlas to axis B) **6th thoracic vertebra to the atlas ✓** C) Sacrum to atlas D) 1st rib to atlas

Exp: Longus colli is a long muscle extending from the 6th dorsal (thoracic) vertebra to the atlas.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the sterno-cephalicus muscle, its two divisions, and their actions.

Quick Recap

- Ventral cervical group: Sterno-cephalicus (2 divisions), Scalenus (dorsal+ventral, can assist inspiration), 3 Rectus capitis muscles (all flex head), Intertransversales colli + Longus colli (flex neck)
- Sterno-cephalicus = unique Spinal Accessory nerve supply
- Sterno-thyro-hyoideus + Omo-hyoideus already covered under Hyoid Region muscles (same muscles, cross-referenced here)

MYOLOGY — NECK: LATERAL CERVICAL GROUP

Lateral Cervical Group — Layers 1-3 (Cross-reference) & Layer 4

Overview — 4 Superimposed Layers

Layer	Muscles
First	Trapezius cervicalis, Brachio-cephalicus
Second	Omo-transversarius, Rhomboideus cervicalis
Third	Serratus cervicis
Fourth	Splenius, Longissimus capitis et atlantis, Complexus (semispinalis capitis), Multifidus cervicis, Obliquus capitis anterior, Obliquus capitis posterior, Rectus capitis dorsalis major, Rectus capitis dorsalis minor

Layers 1-3 — Cross-Reference

- **Trapezius cervicalis + Brachio-cephalicus + Omo-transversarius + Rhomboideus cervicalis + Serratus cervicis:** ALL already fully described under Muscles of the Pectoral Girdle — same muscles, just their "cervicalis"-named portions being placed here in the neck's own layer system

Layer 4 — Detail Table (New Muscles)

Muscle	Origin	Insertion	Action	Blood/Nerve
Splenius	Summits, first 3-4 dorsal spines	Occipital + C1-C3 transverse processes (common with omo-transversarius, cleido-occipitalis, longissimus capitis et atlantis)	Elevate + extend head/neck, or incline toward the neck	Deep cervical artery; Cervical spinal nerves except first 2
Longissimus capitis et atlantis	2 parallel bundles, deep to splenius. Origin: T1-T2 transverse process + cervical vertebrae articular processes (except first 2-3)	Mastoid process + atlas wing	Similar to splenius	Deep cervical + vertebral arteries; Cervical spinal nerves except first 2

Muscle	Origin	Insertion	Action	Blood/Nerve
Complexus (semispinalis capitis)	Spines, T2-T4 + transverse processes, T1-T10 + articular processes, last 5 cervical vertebrae	Occipital bone, lateral to ligamentum nuchae insertion	CHIEF extensor of the head; incline it laterally	Deep cervical + occipital arteries; Cervical spinal nerves
Multifidus cervicis (semispinalis of the neck)	Posterior articular processes, last 5 cervical vertebrae	Spines + posterior articular processes, preceding cervical vertebrae	Extend the head, or flex it laterally	Deep cervical + vertebral arteries; Cervical spinal nerves except first 2
Obliquus capitis anterior (obliquus capitis cranialis)	Interior edge + ventral surface, atlas wing	Paramastoid + mastoid processes	Extend head on the atlas, flex head to a side	Occipital artery; 1st cervical spinal nerve
Obliquus capitis posterior (obliquus capitis caudalis)	Posterior oblique process + spine of the axis	Dorsal face, atlas wing	Rotate the atlas (and head) to one side	Occipital artery; 2nd cervical spinal nerve
Rectus capitis dorsalis major	Spine of the axis	Occipital bone, near external occipital protuberance	Extend the head	Occipital artery; 2nd cervical spinal nerve
Rectus capitis dorsalis minor	Dorsal arch, atlas	Occipital bone, between external occipital protuberance + foramen magnum	Assist the preceding muscle	Occipital artery; 1st cervical spinal nerve

- Splenius: triangular muscle
- Multifidus cervicis: made of oblique bundles between complexus + longissimus capitis et atlantis (external) and lamellar ligamentum nuchae + longissimus dorsi's superior cervical division (internal); lies over T1 + last 5 cervical vertebrae

IMPORTANT

Complexus (semispinalis capitis) is explicitly named the CHIEF extensor of the head — the single most important head-extending muscle in this whole layer-4 group, distinguishing it from the several "assist"-type muscles around it.

EXAM TIP

Obliquus capitis ANTERIOR flexes the head to a side (bends it); Obliquus capitis POSTERIOR rotates the atlas/head to one side (twists it) — similar-sounding names, genuinely DIFFERENT movement types (flexion vs rotation). Don't conflate them.

MNEMONIC

"1st cervical nerve for the SMALL/anterior muscles (ventralis minor, lateralis, obliquus anterior, rectus dorsalis minor); 2nd cervical nerve for the muscles tied to the AXIS (obliquus posterior, rectus dorsalis major)" — nerve level roughly tracks which vertebra (atlas vs axis) the muscle relates to.

PYQ

Which layer-4 lateral cervical muscle is described as the "chief extensor of the head"?

- A) Splenius **B) Complexus (semispinalis capitis) ✓** C) Multifidus cervicis D) Rectus capitis dorsalis minor

Exp: Complexus (semispinalis capitis) is the chief extensor of the head, also inclining it laterally.

PYQ

Obliquus capitis posterior's specific action is to?

A) Flex the head to a side **B) Rotate the atlas and head to one side ✓** C) Extend the head only D) Depress the head

Exp: Obliquus capitis posterior rotates the atlas (and with it the head) to one side, distinct from obliquus capitis anterior's simple lateral flexion.

PYQ

Rectus capitis dorsalis minor's action is described as?

A) Independent head extension **B) Assisting rectus capitis dorsalis major ✓** C) Rotating the atlas D) Flexing the neck laterally

Exp: Rectus capitis dorsalis minor simply assists the (larger) rectus capitis dorsalis major in extending the head.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the complexus muscle and explain why it is considered the chief extensor of the head.

Quick Recap

- Layers 1-3 (Trapezius cervicalis, Brachiocephalicus, Omo-transversarius, Rhomboideus cervicalis, Serratus cervicis) = already covered in Pectoral Girdle module
- Layer 4 = 8 muscles; Complexus = CHIEF head extensor; Obliquus anterior=flexes head sideways, Obliquus posterior=ROTATES head
- Nerve pattern roughly follows atlas-related muscles→1st cervical nerve, axis-related→2nd cervical nerve

MYOLOGY — NECK: HORSE/DOG VARIATIONS

Neck Muscles — Horse Vs Dog Variations

Horse — Key Differences from Ox

Muscle	Horse Variation
Cutaneous colli	PRESENT (unlike the general neck rule that cutaneous muscle is absent — horse has this one exception)
Sterno-cephalicus	ONLY the sterno-mandibularis part is present, representing the whole muscle
Sterno-thyro-hyoideus	Behaves as a DIGASTRIC muscle; hyoid band is LARGER, closely applied to its fellow all the way to insertion
Scalenus dorsalis	LESS extensive
Omo-hyoideus	Also called "subscapulo-hyoideus"; arises from the SUBSCAPULAR fascia

Dog — Key Differences from Ox

Muscle	Dog Variation
Cutaneous muscle	Cervical part IS present
Sterno-cephalicus	Well developed
Omo-hyoideus	ABSENT
Scalenus	Very LONG
Rectus capitis ventralis major	Arises from transverse processes, C2 to C6
Splenius	Extensive

Muscle	Dog Variation
Complexus	Consists of TWO parts

IMPORTANT

Omo-hyoideus shows OPPOSITE presence in horse vs dog — present (renamed "subscapulo-hyoideus") in horse, but completely ABSENT in dog. A clean contrast pair.

PYQ

The omo-hyoideus muscle, also called "subscapulo-hyoideus" and arising from subscapular fascia, is a variation seen in which species?

A) Ox B) Horse ✓ C) Dog D) All three

Exp: In the horse, omo-hyoideus is called subscapulo-hyoideus and arises from the subscapular fascia; it is entirely absent in the dog.

PYQ

"Cutaneous colli," an exception to the general rule that cutaneous muscle is absent in the neck, is present in which species?

A) Ox B) Horse ✓ C) Dog only D) Neither horse nor dog

Exp: The horse has a cutaneous colli muscle, an exception to the general absence of cutaneous muscle in the neck region.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Compare the omo-hyoideus muscle across ox, horse, and dog.

Quick Recap

- Horse: has cutaneous colli (exception to "no cutaneous muscle in neck" rule); omo-hyoideus = "subscapulo-hyoideus" (subscapular fascia origin)
- Dog: omo-hyoideus = ABSENT entirely; scalenus = very long; complexus = 2 parts

MYOLOGY — THORAX

Intercostal & Rib-related Thoracic Muscles

Overview

- Thorax/costal region muscles = attached to thoracic vertebrae, ribs/cartilages, sternum — all RESPIRATORY in function
- List: Serratus thoracis (already covered, Pectoral Girdle module) | Levatores costarum (already covered, Back/Loin module Layer 5) | External intercostals | Internal intercostals | Retractor costarum | Rectus thoracis | Transversus thoracis | Diaphragm

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
External intercostals	Fill intercostal spaces from levatores costarum to the rib's distal extremity; do NOT extend into interchondral spaces; fibres directed DOWNWARD	Posterior border of a rib	Anterior border + external surface, rib behind	INSPIRATORY muscle	Intercostal arteries; Intercostal nerves

Muscle	Description	Origin	Insertion	Action	Blood/Nerve
Internal intercostals	Internal to external intercostals; DO extend into interchondral spaces; fibres directed down-forward	Anterior border, rib + its cartilage (except 1st)	Posterior border, rib in front + its cartilage	Assist EXPIRATION, especially FORCED expiration	Intercostal + internal thoracic arteries; Intercostal nerves
Retractor costarum (retractor of last rib)	Small, triangular; behind the last rib, under cover of serratus dorsalis posterior's last digitation	Transverse processes, first 2-3 lumbar vertebrae	Posterior border of the last rib	Retract the last rib, aid FORCED expiration	1st-4th lumbar arteries; 1st-4th lumbar spinal nerves
Rectus thoracis	Thin, quadrilateral; lateral face, first few ribs + cartilages, under cover of deep pectoral	Lower half, lateral surface, 1st rib	3rd or 4th costal cartilage	Assist INSPIRATION	Internal + external thoracic arteries; First 5 intercostal nerves
Transversus thoracis (triangularis sterni)	Flat, triangular; dorsal sternum face	Sternal ligament	Cartilages, ribs 2-8 + adjacent rib parts	Assist EXPIRATION	Internal thoracic artery; Intercostal nerves

IMPORTANT

External intercostals = INSPIRATORY; Internal intercostals = EXPIRATORY — directly mirroring the same opposite-pair pattern seen with Serratus dorsalis anterior (inspiratory) vs posterior (expiratory) in the back muscles. Recognizing this repeated "external/anterior=in, internal/posterior=out" pattern across multiple muscle pairs makes memorization far easier.

EXAM TIP

Both internal intercostals AND transversus thoracis extend into/across the interchondral (cartilage) region and share the SAME expiratory role — think of them as a coordinated "deep expiratory team."

PYQ

Which of the following IS an inspiratory thoracic muscle?

- A) Internal intercostals B) Transversus thoracis **C) External intercostals ✓** D) Retractor costarum

Exp: External intercostals are inspiratory; internal intercostals, transversus thoracis, and retractor costarum are all involved in expiration (especially forced expiration).

PYQ

The retractor costarum specifically retracts which rib?

- A) The first rib **B) The last rib ✓** C) All ribs equally D) The 8th rib

Exp: Retractor costarum (retractor of the last rib) specifically retracts the last rib, aiding forced expiration.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Differentiate the actions of external and internal intercostal muscles.

Quick Recap

- External intercostals = INSPIRATORY | Internal intercostals + Transversus thoracis + Retractor costarum = EXPIRATORY (esp. forced expiration)

- Rectus thoracis = assists inspiration (small, under deep pectoral, ribs 1-4 region)
- Pattern repeats: "outer/external muscle = inspiratory, inner/internal = expiratory" seen across multiple thoracic/back muscle pairs

MYOLOGY — THORAX

The Diaphragm

Overview

- Broad, LARGE, dome-shaped, UNPAIRED muscle — partition between thoracic + abdominal cavities
- Thoracic surface: convex, covered by pleura | Abdominal surface: concave, covered by peritoneum
- Anteriorly: related to diaphragmatic lung lobes + last 6 ribs (part)
- Posteriorly: related to rumen, spleen, reticulum, liver, omasum, kidneys, adrenals

Structural Parts

Part	Description
Tendinous centre	Aponeurotic middle; partially divided into right + left parts by the crura's descent into it
Muscular rim	= Costal part + Sternal part
Crura (lumbar part)	Right + left crura, each fleshy then tendon-terminated, attached to lumbar vertebrae

Attachments

Part	Attachment Detail
Costal part	Straight line, from the last rib (a hand's breadth above its middle) to the 8th rib's junction + cartilage; consists of a series of digitations
Sternal part	Along the 8th rib's cartilage + across the xiphoid cartilage's abdominal face
Right crus	Attached to the 1st lumbar vertebra's body via the ventral longitudinal ligament; LARGER of the two crura; divides into 2 branches which circumscribe the hiatus oesophagi, unite below, spread to the aponeurotic centre
Left crus	Attached to the FIRST TWO lumbar vertebrae via the ventral longitudinal ligament

Three Foramina (Openings)

Foramen	Location	Transmits
Hiatus aorticus	Between the 2 crura, below T13	Aorta, vena hemiazygos, cisterna chyli
Hiatus oesophagi	Between the right crus's 2 divisions	Oesophagus + dorsal/ventral vagal (oesophageal) continuations
Foramen venae cavae	Slightly below + right of the hiatus oesophagi	Posterior (caudal) vena cava

- Action: CHIEF inspiratory muscle
- Blood supply: Phrenic + musculo-phrenic arteries | Nerve supply: Phrenic nerve

IMPORTANT

The diaphragm is the CHIEF (primary) inspiratory muscle of the entire body — everything else (external intercostals, rectus thoracis, levatores costarum, scalenus when neck fixed) merely ASSISTS inspiration; the diaphragm does the majority of the work.

EXAM TIP

3 foramina, easy to remember by content-type: Hiatus AORTICUS = vascular (aorta+related); Hiatus OESOPHAGI = digestive tube (oesophagus+vagus); Foramen VENAE CAVAE = venous (vena cava). Each name literally tells you what passes through.

Species Differences

Species	Key Feature
Horse	External + internal intercostals THINNER; Transversus thoracis inserted to 7 sternal cartilages (2nd-8th); Diaphragm LESS wide but MORE sloping; right crus does NOT divide into 2 branches; hiatus oesophagi found in the right crus slightly above its aponeurotic-centre junction; foramen venae cavae placed slightly HIGHER than in ox
Dog	Diaphragm STRONGLY curved, aponeurotic centre SMALL; sternal part has the PERICARDIUM attached to it via the pericardio-phrenic ligament; other muscles mostly resemble ox

IMPORTANT

In the dog, the diaphragm's sternal part is directly attached to the PERICARDIUM via the pericardio-phrenic ligament — a distinctive species feature with potential clinical relevance (e.g., diaphragmatic-pericardial hernia is a recognized condition in dogs).

PYQ

The diaphragm is functionally the?

- A) Chief expiratory muscle **B) Chief inspiratory muscle ✓** C) Purely a postural muscle D) An accessory muscle only

Exp: The diaphragm is the chief (primary) inspiratory muscle of the body.

PYQ

The oesophagus and vagal continuations pass through which diaphragmatic opening?

- A) Hiatus aorticus **B) Hiatus oesophagi ✓** C) Foramen venae cavae D) None of these

Exp: The hiatus oesophagi, formed between the two divisions of the right crus, transmits the oesophagus and its vagal continuations.

PYQ

In the dog, which structure is attached to the diaphragm's sternal part via the pericardio-phrenic ligament?

- A) Liver **B) Pericardium ✓** C) Spleen D) Stomach

Exp: In the dog, the pericardium attaches to the diaphragm's sternal part via the pericardio-phrenic ligament.

PYQ

Which crus of the diaphragm is larger and divides into two branches circumscribing the hiatus oesophagi?

- A) Left crus **B) Right crus ✓** C) Both equally D) Neither

Exp: The right crus is larger and divides into two branches that circumscribe the hiatus oesophagi before uniting and spreading to the aponeurotic centre.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the diaphragm's structure, attachments, and the three foramina that pierce it.

Quick Recap

- Diaphragm = unpaired, dome-shaped; Tendinous centre + Muscular rim (costal+sternal) + 2 Crura (lumbar part)
- 3 foramina: Hiatus aorticus (aorta), Hiatus oesophagi (oesophagus+vagus), Foramen venae cavae (caudal vena cava)
- CHIEF inspiratory muscle of the body
- Dog: pericardium attaches to sternal part (pericardio-phrenic ligament) — clinically relevant to diaphragmatic-pericardial hernia

MYOLOGY — TONGUE, SOFT PALATE, PHARYNX & LARYNX

Muscles of the Tongue

Overview — Extrinsic vs Intrinsic

- **Extrinsic muscles:** arise OUTSIDE the tongue (from bone/cartilage), insert into it — control tongue's POSITION (protrusion/retraction/depression)
- **Intrinsic muscles:** bundles running in different directions WITHIN the tongue itself, blending with the extrinsic muscles — control tongue's SHAPE

Extrinsic Muscles — 4 Muscles

Muscle	Origin	Insertion/Course	Action
Stylo-glossus	Great cornu of hyoid bone, near its anterior extremity	Long muscle, ventro-lateral tongue face; terminates near the tip, blending with its fellow + intrinsic muscles	Retract the tongue, draw it to a side
Hyoglossus	Body + root of the tongue (wide, flat muscle)	—	Retract AND depress the tongue
Palatoglossus	Edge of the soft palate	Thin, narrow; passes downward, blends with middle hyoglossus, ends on tongue root	Narrows the isthmus faucium (oral-pharyngeal opening)
Genioglossus	Medial face, mandible ramus, just behind the symphysis	Fan-shaped; fibres pass up to the tongue's body + root	Retract, depress, AND protract the tongue

- Blood supply (all 4): Lingual + sublingual arteries | Nerve supply: Hypoglossal nerve

IMPORTANT

Genioglossus is the ONLY extrinsic tongue muscle that can PROTRACT (push forward) the tongue — the other three only retract/depress/narrow. This asymmetry matters clinically: genioglossus weakness is associated with airway obstruction risk (tongue falls backward) in some species/conditions.

Intrinsic Muscles

- Bundles of muscle fibres running in DIFFERENT directions, forming the bulk of the tongue
- Blend seamlessly with the extrinsic muscles — no sharp anatomical boundary between the two groups

- Function: fine control of tongue SHAPE (curling, flattening, narrowing) during prehension, mastication, and swallowing

EXAM TIP

All 4 extrinsic tongue muscles share the SAME nerve (Hypoglossal, cranial nerve XII) — a clean, easy-to-remember fact, unlike the mixed nerve patterns seen in facial/hyoid muscles.

MNEMONIC

"Stylo retracts+sideways, Hyo retracts+down, Palato narrows the gate, Genio pushes OUT" — one action-word per extrinsic tongue muscle.

PYQ

Which extrinsic tongue muscle is unique in being able to protract the tongue?

- A) Styloglossus B) Hyoglossus C) Palatoglossus **D) Genioglossus ✓**

Exp: Genioglossus retracts, depresses, AND protracts the tongue — the only one of the four capable of protraction.

PYQ

All extrinsic muscles of the tongue are supplied by which cranial nerve?

- A) Glossopharyngeal (IX) B) Vagus (X) **C) Hypoglossal (XII) ✓** D) Facial (VII)

Exp: All four extrinsic tongue muscles (styloglossus, hyoglossus, palatoglossus, genioglossus) are supplied by the hypoglossal nerve.

PYQ

The palatoglossus muscle's specific action is to?

- A) Protract the tongue **B) Narrow the isthmus faucium ✓** C) Depress the tongue only D) Elevate the larynx

Exp: Palatoglossus narrows the isthmus faucium (the opening between oral cavity and pharynx), aiding in swallowing.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Differentiate between extrinsic and intrinsic muscles of the tongue, listing the four extrinsic muscles.

Quick Recap

- Extrinsic (4): Styloglossus (retract+side), Hyoglossus (retract+depress), Palatoglossus (narrow isthmus), Genioglossus (retract+depress+PROTRACT — unique)
- ALL extrinsic muscles = Hypoglossal nerve (XII)
- Intrinsic = shape-control bundles, blend with extrinsic muscles, no sharp boundary

MYOLOGY — TONGUE, SOFT PALATE, PHARYNX & LARYNX

Muscles of the Soft Palate

Overview

- List: Palato-glossus (already covered under Tongue) | Palato-pharyngeus | Tensor palati | Levator palati | Palatinus

Muscle	Description	Action
Palato-Pharyngeus	Placed next to the glandular layer, posterior half of soft palate; continuous with its fellow across the midline; extends down-back to the pharynx's lateral wall, lying external to the posterior pillar, inserts into the thyroid cartilage	Tenses the soft palate, elevates the pharynx, draws larynx + oesophagus toward the tongue root

Muscle	Description	Action
Tensor Palati	Arises from the muscular process of the petrous temporal bone; passes lateral to levator palati; tendon reflects around the pterygoid bone's hamulus, spreads into the soft palate	Tenses the palate
Levator Palati	Origin: muscular process, petrous temporal. Insertion: soft palate	Raises the soft palate toward the pharynx roof; CLOSES the posterior nares during deglutition (swallowing)
Palatinus	Small, elongated, bright red fasciculus; paired, enclosed in the palatine aponeurosis; arises from the bony palate's posterior border, inserts near the soft palate's free border	Shortens the soft palate

- Blood supply: Lesser palatine arteries | Nerve supply: Vagus, Glosso-pharyngeal, AND Trigeminal nerves (mixed supply — unusual, 3 cranial nerves contribute)

IMPORTANT

Levator palati's action of closing the posterior nares during swallowing is CRITICAL — this is exactly what prevents food/liquid from regurgitating up into the nasal cavity during deglutition. Failure of this mechanism (e.g. cleft palate, nerve damage) causes nasal reflux of food, a real clinical sign.

EXAM TIP

"Tensor" vs "Levator" — Tensor palati TENSES (tightens) the palate; Levator palati RAISES it toward the pharynx AND closes the nares. Similar names, distinctly different actions — don't conflate them.

PYQ

Which soft palate muscle's action prevents food from entering the nasal cavity during swallowing?

- A) Tensor palati **B) Levator palati ✓** C) Palatinus D) Palato-pharyngeus

Exp: Levator palati raises the soft palate and closes the posterior nares during deglutition, preventing nasal reflux.

PYQ

The soft palate muscles receive a mixed nerve supply from how many cranial nerves?

- A) 1 B) 2 **C) 3 ✓** D) 4

Exp: Soft palate muscles are supplied by the vagus, glossopharyngeal, AND trigeminal nerves — three cranial nerves contributing together.

PYQ

Tensor palati's tendon reflects around which structure before spreading into the soft palate?

- A) Hyoid bone **B) Hamulus of the pterygoid bone ✓** C) Thyroid cartilage D) Styloid process

Exp: The tensor palati tendon passes lateral to levator palati and reflects around the pterygoid bone's hamulus.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the levator palati muscle and explain its role in preventing nasal regurgitation during swallowing.

Quick Recap

- 5 soft palate muscles (incl. palatoglossus): Palato-pharyngeus (tenses palate, elevates pharynx), Tensor palati (tenses), Levator palati (raises + closes nares — clinically critical), Palatinus (shortens)
- Nerve supply = mixed: Vagus + Glossopharyngeal + Trigeminal

Muscles of the Pharynx

Overview

- Form the muscular wall of the pharynx
- List: Stylo-pharyngeus | Palato-pharyngeus (already covered) | Pterygo-pharyngeus | Aryteno-pharyngeus | Hyo-pharyngeus | Thyro-pharyngeus | Crico-pharyngeus

Muscle	Origin	Action
Stylo-pharyngeus	Great cornu of hyoid bone, near its dorsal extremity	DILATOR of the pharynx (the only dilator in this list!)
Pterygo-pharyngeus	Pterygoid process	Shortens (constricts) the pharynx
Aryteno-pharyngeus	Posterior aspect of the arytenoid cartilage (small muscle)	Constricts the pharynx
Hyo-pharyngeus	Thyroid cornu of the hyoid bone	Constricts the pharynx
Thyro-pharyngeus	Lamina of the thyroid cartilage	Constricts the pharynx
Crico-pharyngeus	Lateral part, arch of the cricoid cartilage	Constricts the pharynx

- Blood supply: Common carotid, external carotid, external maxillary arteries | Nerve supply: 5th (Trigeminal), 9th (Glossopharyngeal), 10th (Vagus) cranial nerves

IMPORTANT

Stylo-pharyngeus is the ONLY dilator among all the pharyngeal muscles — Pterygo-, Aryteno-, Hyo-, Thyro-, and Crico-pharyngeus are ALL constrictors. A classic "odd one out" exam fact — 5 constrictors vs 1 dilator.

EXAM TIP

Notice the naming pattern — each "-pharyngeus" muscle is named for its ORIGIN structure (Stylo=styloid/hyoid, Pterygo=pterygoid, Aryteno=arytenoid, Hyo=hyoid, Thyro=thyroid cartilage, Crico=cricoid cartilage). Once you know this pattern, the origin is embedded in the name itself.

MNEMONIC

"Stylo Dilates, everyone else Constricts" — the simplest way to remember which of the 6 pharyngeal muscles breaks the pattern.

PYQ

Which pharyngeal muscle uniquely acts as a dilator, while all others constrict?

- A) Crico-pharyngeus **B) Stylo-pharyngeus ✓** C) Thyro-pharyngeus D) Hyo-pharyngeus

Exp: Stylo-pharyngeus is the sole dilator of the pharynx; all five other pharyngeal muscles are constrictors.

PYQ

The pharyngeal muscles are supplied by cranial nerves numbered?

- A) 3, 4, 6 **B) 5, 9, 10 ✓** C) 7, 8, 11 D) 2, 5, 9

Exp: Pharyngeal muscles receive supply from the 5th (trigeminal), 9th (glossopharyngeal), and 10th (vagus) cranial nerves.

PYQ

Crico-pharyngeus arises from which structure?

- A) Thyroid cartilage lamina B) Arytenoid cartilage **C) Lateral arch of the cricoid cartilage ✓** D) Hyoid bone's thyroid cornu

Exp: Crico-pharyngeus arises from the lateral part of the cricoid cartilage's arch.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

List the muscles of the pharynx, identifying which one acts as a dilator.

Quick Recap

- 6 pharyngeal muscles named by origin structure: Stylo-, Pterygo-, Aryteno-, Hyo-, Thyro-, Crico-pharyngeus (+ Palato-pharyngeus already covered)
- ONLY Stylo-pharyngeus DILATES; all others CONSTRICT
- Nerve supply: cranial nerves 5, 9, 10

MYOLOGY — TONGUE, SOFT PALATE, PHARYNX & LARYNX

Muscles of the Larynx

Overview — Extrinsic vs Intrinsic

Group	Muscles
Extrinsic	Sterno-thyro-hyoideus (already covered under Hyoid Region), Thyro-hyoideus, Hyo-epiglotticus
Intrinsic	Crico-thyroideus, Crico-arytenoideus dorsalis, Crico-arytenoideus lateralis, Arytenoideus transversus, Thyro-arytenoideus

Extrinsic Muscles

Muscle	Course	Action	Blood/Nerve
Thyro-hyoideus (Hyothyroideus)	Extends between the thyroid cartilage and the hyoid body + thyroid cornua	Draws the larynx downward and forward	Anterior laryngeal artery; Anterior laryngeal branch of vagus
Hyo-epiglotticus	Small, BIFID muscle, enclosed in the glosso-epiglottic fold; arises from hyoid body + 2 small cornua; inserts on the ventral half, anterior epiglottis face	Draws the epiglottis toward the tongue root	Lingual artery; Hypoglossal nerve

Intrinsic Muscles — The Core Larynx Movers

Muscle	Location/Course	Action	Blood/Nerve
Crico-thyroideus	Posterior part, lateral larynx surface; cricoid cartilage attachment to thyroid cartilage's posterior cornu	Draws thyroid + ventral cricoid cartilage TOGETHER — TENSES the vocal folds	Posterior laryngeal artery; Anterior laryngeal nerve
Crico-arytenoideus dorsalis	Dorsal cricoid surface; cricoid lamina to arytenoid's muscular process	DILATES the glottis; ABDUCTS the vocal process + fold	Posterior laryngeal artery; Posterior laryngeal nerve
Crico-arytenoideus lateralis	Deep thyroid cartilage surface; upper cricoid to arytenoid's muscular process	CLOSES the glottis by rotating the arytenoid cartilage MEDIALY	Posterior thyroid artery; Posterior laryngeal nerve

Muscle	Location/Course	Action	Blood/Nerve
Arytenoideus transversus	UNPAIRED; dorsal face, arytenoid cartilage bodies, between their lateral borders	NARROWS the glottis	Posterior laryngeal artery; Posterior laryngeal nerve
Thyro-arytenoideus	Medial thyroid cartilage lamina to the arytenoid's muscular process	CLOSES the glottis; SLACKENS the vocal folds	Same as preceding

IMPORTANT

Crico-arytenoideus DORSALIS is the ONLY glottis-DILATING muscle in the entire larynx — every other intrinsic muscle either closes/narrows the glottis or tenses/slackens the vocal folds. This single muscle is critically important: its paralysis (common in "roaring" / laryngeal hemiplegia in horses) causes airway obstruction because nothing can open the glottis properly during exercise.

EXAM TIP

Crico-thyroideus TENSES the vocal folds (tightens, for pitch); Thyro-arytenoideus SLACKENS them (loosens) — these two muscles work as an opposing PAIR controlling vocal fold tension, similar to how flexor/extensor pairs work at a joint.

CLINICAL PEARL

Crico-arytenoideus dorsalis paralysis is the anatomical basis of "roaring" (recurrent laryngeal neuropathy) in horses — a real, commonly tested clinical correlation in equine medicine, since this is the sole glottis dilator and its failure directly causes exercise intolerance/airway noise.

MNEMONIC

"Dorsalis Dilates (the only one), Lateralis + Thyro-arytenoideus close, Transversus narrows, Crico-thyroideus tenses" — one clause per intrinsic muscle's core action.

PYQ

Which intrinsic laryngeal muscle is the sole dilator of the glottis?

- A) Crico-thyroideus **B) Crico-arytenoideus dorsalis ✓** C) Crico-arytenoideus lateralis D) Thyro-arytenoideus

Exp: Crico-arytenoideus dorsalis is the only glottis-dilating muscle; all other intrinsic laryngeal muscles close/narrow the glottis or adjust vocal fold tension.

PYQ

Paralysis of which muscle is the anatomical basis of "roaring" (laryngeal hemiplegia) in horses?

- A) Thyro-arytenoideus B) Crico-thyroideus **C) Crico-arytenoideus dorsalis ✓** D) Arytenoideus transversus

Exp: Crico-arytenoideus dorsalis paralysis prevents glottis dilation, causing the airway obstruction seen in equine laryngeal hemiplegia ("roaring").

PYQ

Which intrinsic laryngeal muscle tenses the vocal folds by drawing the thyroid and cricoid cartilages together?

- A) Thyro-arytenoideus **B) Crico-thyroideus ✓** C) Crico-arytenoideus lateralis D) Arytenoideus transversus

Exp: Crico-thyroideus tenses the vocal folds by approximating the thyroid cartilage and the ventral cricoid cartilage.

PYQ

Which extrinsic laryngeal muscle draws the epiglottis toward the root of the tongue?

- A) Thyro-hyoideus B) Sterno-thyro-hyoideus **C) Hyo-epiglotticus ✓** D) Crico-thyroideus

Exp: Hyo-epiglotticus, a small bifid muscle in the glosso-epiglottic fold, draws the epiglottis toward the tongue root.

PYQ

The arytenoideus transversus muscle is unique among intrinsic laryngeal muscles because it is?

A) The only dilator **B) Unpaired ✓** C) Supplied by the hypoglossal nerve D) Attached to the hyoid bone

Exp: Arytenoideus transversus is unpaired, unlike the other intrinsic laryngeal muscles which are paired structures.

THEORY PYQ · Asked in: KVAFSU 1st Prof Anatomy

Describe the intrinsic muscles of the larynx and explain the clinical significance of crico-arytenoideus dorsalis.

Quick Recap

- Extrinsic: Thyro-hyoideus (draws larynx down-forward), Hyo-epiglotticus (pulls epiglottis to tongue root)
 - Intrinsic: Crico-thyroideus (TENSES folds), Crico-arytenoideus dorsalis (ONLY dilator — "roaring" in horses), Crico-arytenoideus lateralis + Thyro-arytenoideus (CLOSE glottis), Arytenoideus transversus (narrows, unpaired)
 - Crico-arytenoideus dorsalis = highest clinical-relevance muscle in this module (equine laryngeal hemiplegia)
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ONLYVET